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1 Supplemental Methods

1.1 Prophylaxis and Empiric Treatment

No oral decontamination or antibacterial prophylaxis was administered peri-HCT. Trimethoprim-sulfamethoxazole (TMP-SMZ) was the preferred *Pneumocystis jiroveci* Pneumonia (PJP) prophylaxis. Micafungin (during conditioning), followed by Voriconazole (on day 2 post HCT) was used for fungal prophylaxis. Patients received acyclovir prophylaxis for first 30 days if HSV serology was positive. First line empiric treatment for fever and neutropenia was cefepime alone unless there was history of multi-drug resistant organism or signs of sepsis. IVIG was given weekly for levels of IgG <400mg/dL. Viral reactivation (e.g. Cytomegalovirus (CMV), Epstein-Barr Virus (EBV), Adenovirus (AdV)) was monitored weekly for first 100 days and treated preemptively with antivirals as appropriate.

1.2 Eligibility Criteria

Eligibility criteria (figure SF1) included all patients less than 19 years of age who received a hematopoietic-cell transplantation at St Jude Children's Research Hospital between 2016 and 2019. Patients were excluded from a time point if there stool samples were missing, had insufficient bacterial content for sequencing or insufficient sequence depth or did not achieve neutrophil recovery.

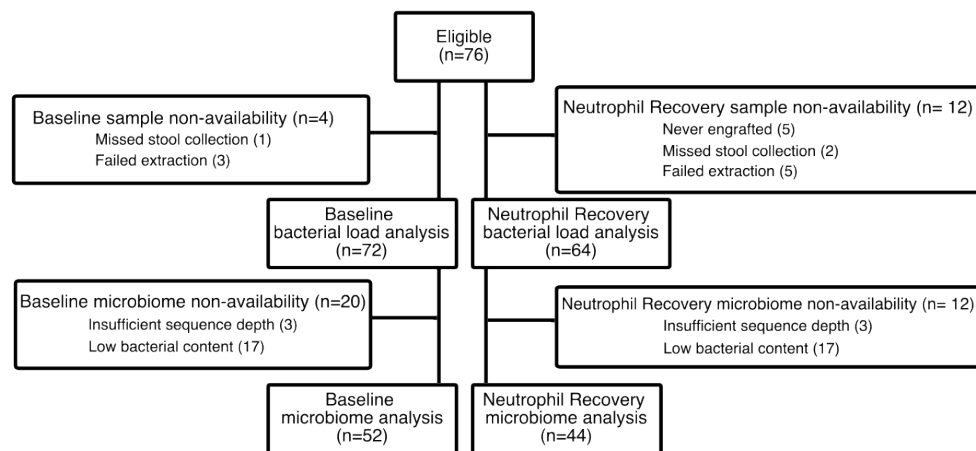


Figure SF1: Sample availability for baseline and neutrophil recovery.

1.3 Sample preparation, DNA extraction, Amplicon preparation, DNA sequencing

Stool samples frozen at -80°C were diluted to 10% w/v with nuclease-free PBS and 20 mg frozen (wet weight) stool equivalent was extracted (modified Qiagen DNeasy PowerSoil 96 HTP (Cat. No. 1115096)) using enzyme treatment (10 µL of Lysozyme (10 mg/ml), 6 µL Mutanolysin (25 kU/ml), 3 µL Lyostaphin (4KU/ml) for 1 hour) and bead-beating (FastPrep) in a protocol optimized for low bacterial content stool samples[1]. Bacterial DNA content was measured using a 16S rRNA quantitative PCR using a plasmid containing *E. coli* 16S gene as the standard, primers and probes as previously described [2] with only change being reverse primer was changed to GGACTACHVGGGTATCTAATCCTGTT to achieve greater universality based on primers from Herlemann et al [3]. Samples with less than 1000 copies of 16S rRNA/mg wet weight of stool were excluded from study. The 16S rRNA gene amplicon for sequencing used primers for the V3-V4 region[4] and a touch-down PCR protocol[5]. The PCR had 20 cycles with annealing step decreasing by 0.3°C each cycle (denaturation 95°C x 20 sec, annealing range of 60-54.3°C x 15 sec and extension 72°C x 5min) followed by 20 cycles with the standard cycling conditions (denaturation 95°C x 20 sec, annealing 55°C x 15 sec and extension 72°C x 5min). Sequencing of indexed amplicons was performed

on an Illumina MiSeq instrument with miseq reagent kit v3-600 cycle to capture paired end (2x 300bp) sequences. For shotgun metagenome sequencing whole-genome libraries were prepared from DNA extracted as above from stool samples and sequencing was performed on an Illumina NovaSeq instrument with NextSeq reagent kit to obtain 100 bp paired-end reads.

1.4 Classification

Paired-end reads were quality filtered, derplicated and clustered into amplicon sequence variants in DADA2 (ver 1.12.0)[6]. The filtered and paired 16S rRNA gene amplicons were used to recruit full-length 16S rRNA gene sequences from Ribosomal Database Project release 16 (March 2018)[7] using vsearch (ver 2.7.1) with global alignment search. The collected reference sequences were assembled into a phylogenetic tree using RaxML (version 8.2.11)[8]; leaves with only 1 representative sequence were used to recruit additional sequences. QC-filtered amplicon sequences were placed onto the classification reference tree using the pplacer tool, using MALIAMPLI[9], with the pipeline[10] validated in separate studies. To ensure that classified taxonomies were not overcalled we clustered sequence variants in terms of phylotypes based on phylogenetic distance and coabundance[9]. Note MALIAMPLI allows with harmonization of classification across microbiome studies so these results can be compared to previous or future cohorts[9].

1.5 Stratification of Microbiome Metrics

Various indices that describe microbiome community structure have been used to illustrate how microbiome disruptions are associated with outcomes in adult HCT studies[11, 12, 13]. All of these indices have distinct distributions in pediatric gut microbiotas than adults, therefore we have chosen to stratify based on healthy pediatric microbiota and classify based on a percentile normalized by age. The healthy pediatric microbiota cohort were from de-identified non-research samples (designated so by St Jude and Seattle Children’s Institutional Review Boards) obtained as leftover control samples (healthy controls that had had no gastrointestinal symptoms for the previous 4 days) from a previous studies. The number of healthy controls in the ≤ 2 yo group was 124 and in the 2–19 yo group was 229. Diversity is a measure of both the evenness and richness of bacterial taxons within a community. Many adult HCT studies use Simpsons diversity index[14, 13], we opted for a phylogenetic diversity as it is less prone to error from over-classification, has grounding in classical phylogenetic diversity (improves interpretation of whether differences are biologically relevant/significant) and provides more discrimination than OTU methods or sequence variation richness. Given that taxonomies have inconsistent phylogenetic relationships in bacteria we have elected to use Balanced Weighted Phylogenetic Diversity (BWPD) to measure diversity in our samples. We found that phylogenetic diversity differed significantly between age categories less than 2 years of age, 2-6 years of age and greater than 5 years of age in healthy pediatric cohort (Figure 1). We opted to stratify based on the 20th percentile healthy cohorts BWPD per age category (ST1).

Low bacterial load is the most common reason samples from HCT are excluded from microbiome studies. To determine if bacterial load, a potential indicator of disturbance, was associated with outcomes we mea-

Metric	<2 yo	2–19 yo	2–5 yo	5–19 yo	Healthy Percentile*	Adult Cutoff
Phylogenetic Diversity	2.98		2.93	2.74	20th	50th percentile HCT cohort [13]
Bacterial Load	1×10^6	9.5×10^5			20th	
Dominant Taxa	79 %	39.1%			90th	30% [12]
Butyrate Producer Relative Abundance	0.5%	6.69%			50th	high (>1%); low (0.01-1%)[11]

Table ST1: Microbiome indices varied per age categories in healthy and HCT pediatric cohorts. The values that correspond to the cutoff percentiles(*) for age matched based on a healthy control cohort are listed and for comparison illustrative example cutoffs from adult microbiome-HCT studies.

sured the 16S rRNA gene content by quantitative PCR, as described in [2]. We found that infants (under the age of 2) had on average higher stool bacterial load in healthy cohort (figure 1), therefore we opted for the twentieth percentile of healthy cohort bacterial load per age category (ST1).

Butyrate production is one of the key links to microbiome metabolism and host response [15, 16]. Studies have linked butyrate production to viral LRTI [11] and recovery from GVHD [17] in adult HCT. Here we infer the potential to produce butyrate based on relative frequency of known butyrate producers [18]. Known butyrate producers are more common in childhood microbiota (2-19 year olds) than that found in adults or in infants (less than 2 years old), therefore we opted to stratify based on the 50th percentile of healthy cohorts butyrogens per age category (ST1).

Taxon dominance is a measure of evenness of bacterial taxons within a community that looks only at the most abundant taxons relative abundance and has been associated with outcomes in adult HCT studies [12, 19]. We found that single taxon dominance is quite common in healthy infant microbiota (figure ??), with dominance differing significantly between age categories less than 2 years of age and greater than 2 years of age. We opted to stratify based on the 90th percentile of healthy cohorts dominance per age category (ST1).

1.6 Composition Attributes and Phylotype Ratios

As composition can vary widely between individuals within a group and important taxa may be absent in various communities, we screened for phylotypes that persistently vary in distribution and abundance between outcome groups identified by Corncob release 0.2.0 (Count Regression for Correlated Observations with the Beta binomial) [20] with a false discovery rate of 0.00001. Screened phylotypes were included in preliminary and final composition attribute ratios if variance persisted when outliers excluded (modified mean). Composition ratios of absolute abundances were constructed from screened phylotypes, with excess zeros of phylotype abundances handled with small positive constant (1) before performing read count ratio and log transformations similar in approach to ANCOM [21]. Ratios that resulted in describing similar taxonomies were discarded in favor of simpler ratios that had fewer total phylotypes. The following (table ST2) is a complete list of the ratios that were considered in this study (all ratios were tested against all outcomes), followed by the sequence variants that made up each of the phylotypes considered.

1.7 Functional Metagenomics

To provide insight into whether the phylotype ratios had functional underpinnings we profiled metabolic pathways (using HUMAnN 3.0) whose presence differed between the denominator and numerator groupings for the ratios that had significant hazards ratios for infectious or acute GVHD outcomes. This filtering was based on absence and likelihood of presence in the genomes for the type strains of each of the taxa represented in the ratios. Pathway coverage was determined based on number of genomes where it was present (1) or had a confidence score less than 1 summed over total number of genomes in that grouping. Once a pathway was designated to differ in a ratio then we assessed Cox proportional-hazards multivariable regression model on the normalized pathway abundance for the outcome of interest (based on association with ratio). Pathway abundance is proportional to the number of complete 'copies' of the pathway in the sample normalized for million total reads (sequencing depth).

1.8 Specific Infection Definitions

Bloodstream infection was defined as detection of bacteria (either recognized bacterial pathogen [22] from or ≥ 1 blood culture or common commensal bacteria if identified from ≥ 2 blood culture sets) or fungi in blood cultures obtained for clinical care. Viral enterocolitis was defined as detection of adenovirus, rotavirus or norovirus by PCR in stool sample obtained with documented clinical diagnosis of diarrhea (>3 loose or watery stools in a 24-hour period). *Clostridioides difficile* infection was defined as identification of toxin gene PCR from loose or watery stool. Parasitic enterocolitis was diagnosed by presence of *Cryptosporidium*

Name	Group	Phylotypes	Taxon Represented
ratio 3.3	17	pt__00189, pt__00421, pt__00235	Abiotrophia defectiva + Enterococcus raffinosus + Rothia aeria + Rothia dentocariosa
	18	pt__00018, pt__00009, pt__00016, pt__00074, pt__00111, pt__00023, pt__00027, pt__00019, pt__00228	Lachnospiraceae + Lacticaseibacillus rhamnosus + Lactobacillus casei group + Lactobacillus casei+ Clostridiales + Streptococcus parasanguinis + Streptococcus australis + Ruminococcus + Ruminococcus faecis + Faecalimonas umbilicata + Lachnospiraceae + Gluceraibacter + Schaalia odontolytica + Actinomyces + Schaalia+ Akkermansia muciniphila + Anaerostipes + Anaerostipes butyraticus
ratio 3.16	11	pt__00189, pt__00421, pt__00235, pt__00001	Abiotrophia defectiva + Enterococcus raffinosus + Rothia aeria + Rothia dentocariosa+ Veillonella parvula + Veillonella rogosae + Veillonella dispar + Veillonella + Veillonella atypica+ Veillonella tobetsuensis + Veillonella denticariosi
	12	pt__00043, pt__00003, pt__00024 , pt__00068, pt__00191, pt__00015	Dorea longicatena + Dorea + Romboutsia timonensis + Romboutsia + Anaerobutyricum + Anaerobutyricum hallii + Limosilactobacillus reuteri + Limosilactobacillus vaginalis + Turicibacter sanguinis + Turicibacter
ratio 5.0	13	pt__00005, pt__00021	Klebsiella pneumoniae+ Ruminococcus gnavus
	14	pt__00025, pt__00018 , pt__00057, pt__00138, pt__00268, pt__00097, pt__00015, pt__00001, pt__00024, pt__00200, pt__00031 , pt__00054, pt__00067, pt__00198, pt__00086, pt__00065, pt__00079 , pt__00042	Lapidilactobacillus dextrinicus+ Limosilactobacillus fermentum+ Lactobacillus gasseri+ Lactobacillus helveticus+ Lacticaseibacillus rhamnosus+ Ligilactobacillus salivarius+ Turcibacter sanguinis+ Veillonella dispar+ Anaerobutyricum hallii+ Lancefieldella parvula+ Bacteroides xylanisolvens+ Bacteroides ovatus+ Phocaeicola vulgatus+ Bacteroides ovatus xylanisolvens+ Clostridium butyricum+ Clostridium perfringens+ Clostridium scindens+ Clostridium subterminale+ Clostridium cadaveris+ Clostridium symbiosum
ratio 7.8	3	pt__00076, pt__00033 , pt__00216, pt__00086, pt__00049, pt__00652	Ruthenibacterium lactatiformans + Enterocloster clostridioforme + Enterocloster boltea + Clostridium + Clostridium cadaveris + Clostridium paraputrificum + Lactobacillus delbrueckii
	4	pt__00421, pt__00263, pt__00706, pt__00179 , pt__00237, pt__00191, pt__00676, pt__00124, pt__00201	Enterococcus raffinosus + Staphylococcus cohnii + Streptococcus anginosus + Parabacteroides merdae + Parabacteroides johnsonii + Turcibacter + Limosilactobacillus vaginalis + Clostridium + Anaerofustis + Anaerofustis stercorihominis + Anaerotignum lactatifermentans + Anaerotignum
ratio 9.0	10	pt__00046, pt__00482	Prevotella + Staphylococcus
	9	pt__00273, pt__00630, pt__00175, pt__00649, pt__00020, pt__00005	Enterobacter+ Citrobacter+ Escherichia+ Escherichia/Shigella + Klebsiella
ratio 11.26	1	pt__00235, pt__00193, pt__00301, pt__00127, pt__00068 , pt__00180	Rothia aeria + Rothia dentocariosa + Veillonella seminalis + Veillonella + Oribacterium sinus + Fusobacterium nucleatum + Fusobacterium periodonticum + Limosilactobacillus reuteri + Leuconostoc mesenteroides + Leuconostoc pseudomesenteroides
	2	pt__0157, pt__00093, pt__00450, pt__00395	Anaerostipes + Anaerostipes butyraticus + Dorea formicigenerans + Agathobaculum + Robinsoniella peoriensis
ratio 12.0	15	pt__00163, pt__00117, pt__00022	Enterococcus avium+ Enterococcus dispar+ Streptococcus sanguinis
	16	pt__00031, pt__00024, pt__00053, pt__00038, pt__00057, pt__00068, pt__00097	Anaerobutyricum hallii+ Bacteroides stercoris+ Phocaeicola vulgatus + Blautia massiliensis+ Lactobacillus helveticus+ Limosilactobacillus reuteri+ Ligilactobacillus salivarius
ratio 16.16	5	pt__00234, pt__00542	Peptoniphilus harei + Peptoniphilus asaccharolyticus
	6	pt__00086, pt__00131, pt__00264, pt__00836, pt__00047, pt__00228, pt__00285	Clostridium + Clostridium cadaveris + Lactobacillus acidiphilus + Catenibacillus + Caproiciproducens + Lacticaseibacillus paracasei + Lacticaseibacillus casei + Lacticaseibacillus casei group + Anaerostipes + Anaerostipes butyraticus +
ratio 18.0	7	pt__00002, pt__00039, pt__00163, pt__00007, pt__00190, pt__00009	Enterococcus avium + Enterococcus casseliflavus + Enterococcus faecalis+ Streptococcus sanguinis+ Streptococcus equinus or infantarius or australis or rubneri
	8	pt__00038, pt__00142	Blautia glucerasea + Blautia massiliensis
ratio 25.26	19	pt__00054, pt__00304, pt__00288	Bacteroides ovatus + Bacteroides xylanisolvens + Schaalia+ Schaalia turicensis+ Actinomyces
	20	pt__0157, pt__00625, pt__00093, pt__00450, pt__00395	Anaerostipes + Anaerostipes butyraticus + Intestinimonas + Dorea formicigenerans + Agathobaculum + Robinsoniella peoriensis

Table ST2: Phylotype ratios include compositional attributes that differed in variance and abundance between groups. All 10 ratios were investigated for their ability to predict infectious and gvhd outcomes in pediatric HCT participants from gut microbiota at baseline and neutrophil recovery.

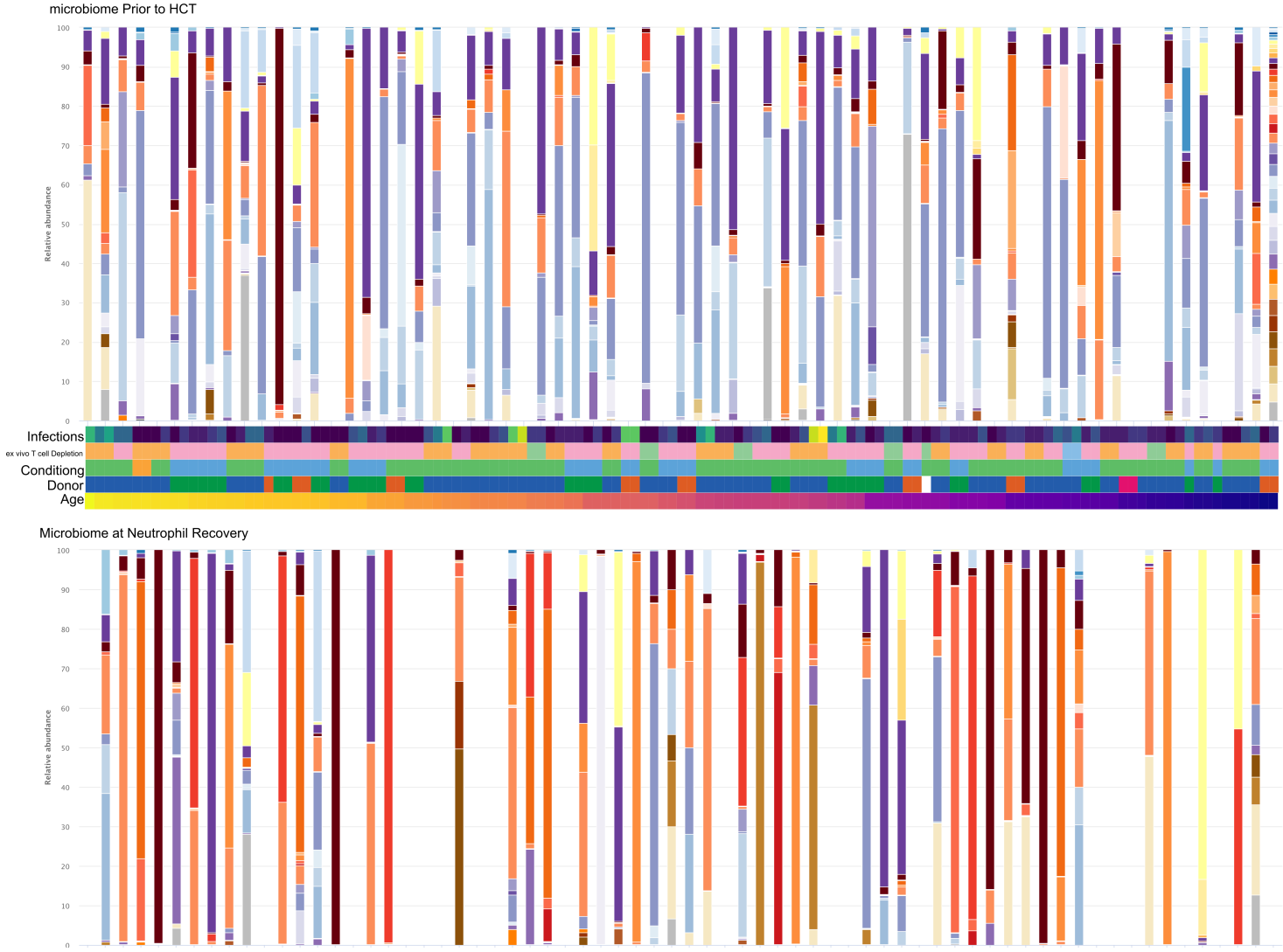
antigen in loose or watery stool. Lower respiratory tract infection was defined as either evidence of infection from bronchoalveolar lavage sample (e.g. positive culture, aspergillus antigen >0.5 or positive viral PCR) or new possible or definite pulmonary infiltrate found in chest imaging (as assessed by radiologist blinded to microbiome and microbiology results of the participants) with positive upper respiratory tract sample (NP swab or tracheal aspirate)[11]. Viremia was defined as the presence of virus in blood stream that would meet treatment standards for Adenovirus (quantitative PCR above 3 log in first four months after HCT), CMV (quantitative PCR above 3 log), EBV (quantitative PCR above 4 log or rising in 3 consecutive weekly tests) or BK virus (requires quantitative PCR above 4 log in blood and above 7 log in urine)[23]. Recurrent episodes of any infection was only considered if occurred more than 14 days after initial positive bacteria culture and more than 90 days for viral infections. Only exception was no recurrence included for reactivation of systemic viruses (e.g. CMV, HHV6, EBV viremia).

1.9 Statistical Methods

Statistical analyses were performed using R version 4.1.0, including R packages (tidyverse, data.table, MASS, stats, ggpubr, corncob). Primary statistical analysis between clinical outcomes and microbial ratios or indices was Cox proportional-hazards multivariable regression model performed using R package survival (coxph). When baseline samples were analyzed, outcomes were considered for all events from HCT day 0. When neutrophil engraftment samples were analyzed, outcomes from neutrophil engraftment were included (did not include day 0 to neutrophil engraftment events). All data was censored at one years of follow-up. For infection outcomes, adjustments in multivariable models were made for age, gender and conditioning regimen (myeloablative or reduced intensity/non-myeloablative). For acute GVHD, additional adjustments in multivariable models were made for graft manipulation (T-cell depletion) and graft source (apheresis or bone marrow). When community indices was considered as a binary (high v low) variable, patients were split into groups based on a percentile compared to healthy pediatric microbiota and normalized by age. We used the Rtsne package to visualize microbiota composition based on t-distributed stochastic neighbor embedding (t-SNE) dimensionality reduction. The classification down to a species level was used as input to the t-SNE algorithm.

2 Supplemental Data

2.1 Microbiome Composition



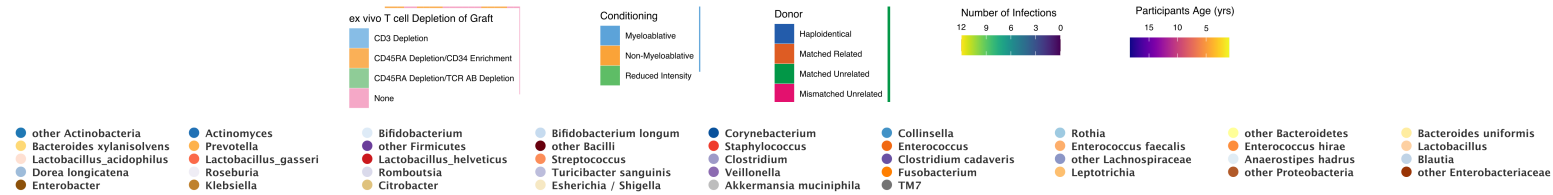


Figure SF2: Microbiome composition in 74 pediatric participants prior to HCT (above) and at neutrophil recovery (below) with each microbiome a bar displayed as relative abundance colors correspond to particular species or if not specified as other phylum/family. Participants age, donor source, conditioning regimen, T cell depletion of graft and number of infections are shown in heat map between the two microbiome charts. Participants are ordered by age on the x-axis and if available both time points are lined up.

Indices	Category	Baseline			Neutrophil Recovery			
		No MDI, Number (%)	MDI, Number (%)	Fischer's Exact (P Value)	No MDI, Number (%)	MDI, Number (%)	Fischer's Exact (P Value)	
Bacterial Load	high	7 (58.3)	34 (56.7)	>0.999	10 (38.5)	14 (36.8)	>0.999	
	low	5 (41.7)	26 (43.3)		16 (61.5)	24 (63.2)		
Diversity	high	4 (40.0)	17 (40.5)	>0.999	6 (35.3)	10 (40.0)	>0.999	
	low	6 (60.0)	25 (59.5)		11 (64.7)	15 (60.0)		
Single Taxa Dominance	present	5(50.0)	12 (26.7)	0.255	13 (65.0)	21 (72.4)	0.754	
	absent	5 (50.0)	33 (73.3)		7 (35.0)	8 (27.6)		
Butyrate Frequency	high	6(60.0)	30(71.4)	0.475	9(52.9)	4(16.0)	0.018	
	low	4(40.0)	12(28.6)		8(47.1)	21(84.0)		

Table ST3: No differences observed in microbiome community indices and number of microbiologically defined infections prior to HCT (baseline) and at the time of neutrophil recovery in pediatric participants. Fischer's exact p value is adjusted for multiple comparisons.

2.2 Community Indices and Number of Microbiologically Defined Infections

Gut microbiota disruption have been linked to clinical outcomes after adult HCT. The main objective of this project was to evaluate whether the same microbiota disruptions or signatures were associated with risk of infection in pediatric HCT. We investigated four signatures of disruption (i.e. Bacterial Load, Diversity, Single Taxa Dominance and Butyrate Producers Relative Frequency) and found that none were associated with having a Microbiologically Defined Infection (MDI) after HCT in pediatric samples taken prior to HCT (baseline) and at the time of neutrophil recovery (Table ST3). Further we could find no association between a particular composition cluster, these signatures of disruption, or age with the number of MDIs that occurred (Figure SF3).

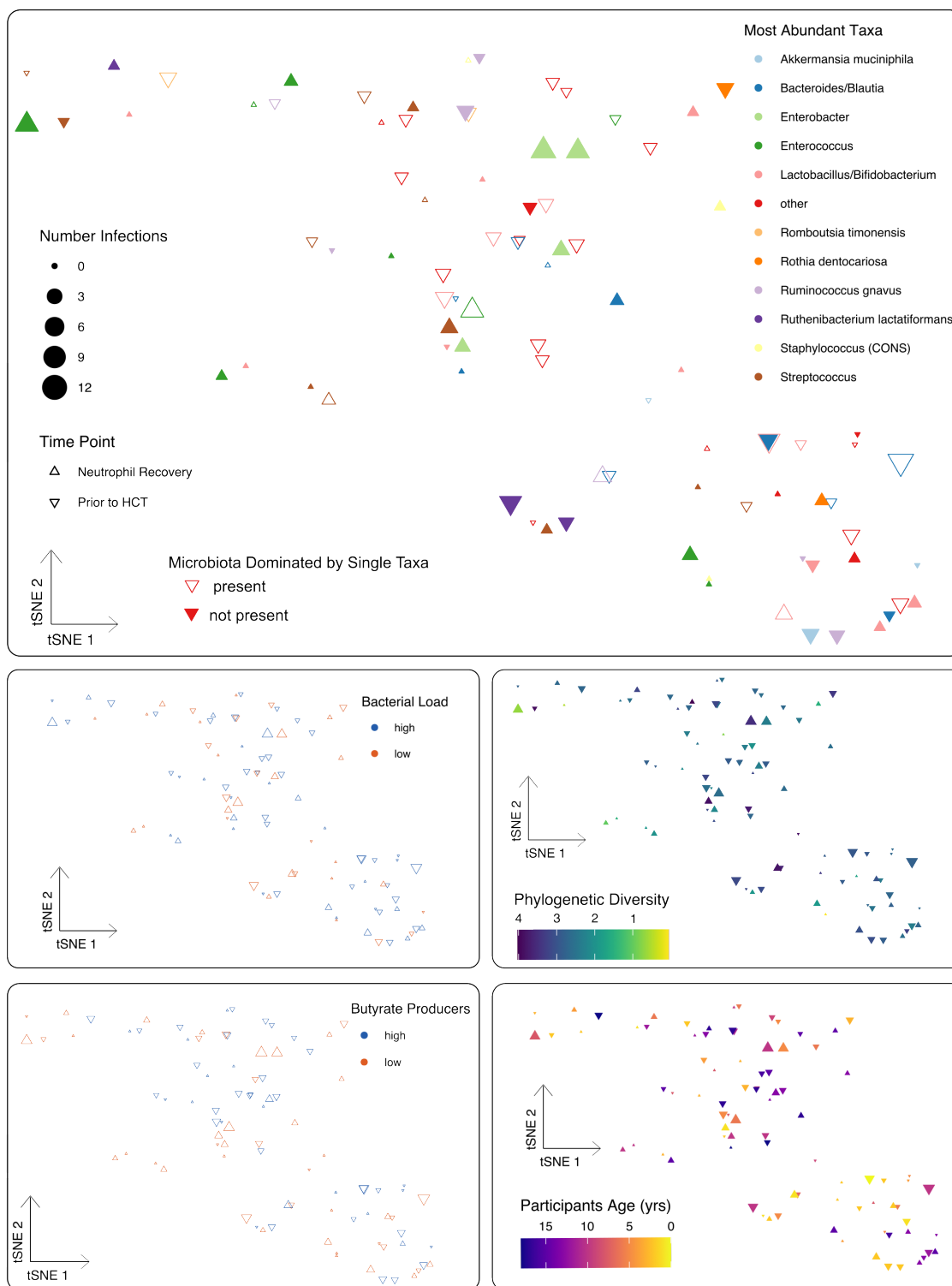


Figure SF3: Microbiome disturbance does not predict infections in pediatric HCT. Microbiomes from 74 pediatric patients prior to HCT and at Neutrophil Recovery are displayed according to their similarity (Kantorovich-Rubinstein distance). The number of microbiologically defined infections determines size. Each triangle represents a single stool sample shown according to the t-distributed stochastic neighbor embedding (t-SNE) algorithm. The axes have arbitrary units with more similar samples clustering together. In upper figure, color corresponds to most abundant taxonomy with filled in triangles signifying that the most abundant taxonomy was dominant (above 90th percentile for age matched controls). In lower figures, color corresponds to either bacterial load (low is below 20th percentile for age matched controls), quantitative phylogenetic diversity, butyrate producers (high above 10th median for age matched controls) or participants age.

2.3 Cox Regression Analysis- Indices, Ratios and Infection Outcomes

Prior to HCT variable	Bacterial Infections			Viral Infections			Viral Enterocolitis			CDI			Bacteremia		
	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value
Bacterial Load (categorical 20th percentile)	1.03	[0.54;1.97]	0.917	0.69	[0.38;1.28]	0.239	0.67	[0.34;1.36]	0.27	0.88	[0.39;1.98]	0.762	1.27	[0.54;2.97]	0.588
Butyrate Producers (categorical 50th percentile)	1.27	[0.55;2.93]	0.580	0.74	[0.34;1.61]	0.454	0.59	[0.24;1.50]	0.269	2.88	[0.81;10.21]	0.101	1.01	[0.31;3.29]	0.991
Phylogenetic Diversity (categorical 20th percentile)	0.84	[0.37;1.94]	0.686	0.93	[0.43;2.02]	0.85	1.21	[0.50;2.96]	0.676	0.49	[0.17;1.42]	0.189	2.67	[0.78;9.14]	0.117
Single Dominant Taxa (categorical 90th percentile)	0.62	[0.26;1.45]	0.271	1.22	[0.57;2.60]	0.603	1.10	[0.45;2.65]	0.840	0.14	[0.03;0.64]	0.011	1.29	[0.44;3.80]	0.650
ratio 9.0 (group 10/ group 9)	1	[0.79;1.26]	0.974	1.16	[0.93;1.44]	0.186	1.09	[0.85;1.39]	0.512	1.01	[0.74;1.37]	0.965	0.97	[0.70;1.33]	0.849
ratio 12.0 (group 15/ group 16)	0.94	[0.76;1.16]	0.574	1.24	[1.04;1.49]	0.018	1.04	[0.83;1.30]	0.756	0.82	[0.62;1.09]	0.164	1.15	[0.87;1.53]	0.314
ratio 18.0 (group 7/ group 8)	1.03	[0.84;1.25]	0.806	1.72	[1.32;2.23]	<0.001	1.32	[1.03;1.70]	0.029	1.03	[0.79;1.36]	0.816	0.95	[0.73;1.23]	0.68
ratio 5.0 (group 13/ group 14)	0.77	[0.59;0.99]	0.044	1.01	[0.81;1.27]	0.911	1.18	[0.91;1.53]	0.209	0.67	[0.47;0.96]	0.029	0.97	[0.71;1.34]	0.867
ratio 11.26 (group 1/ group 2)	1.72	[1.25;2.37]	<0.001	1.39	[1.04;1.85]	0.025	1.28	[0.94;1.73]	0.113	1.65	[1.13;2.41]	0.009	1.24	[0.86;1.80]	0.251
ratio 16.16 (group 5/ group 6)	1.11	[0.86;1.44]	0.426	1.66	[1.24;2.23]	<0.001	2.76	[1.75;4.36]	<0.001	1.23	[0.89;1.70]	0.209	1.08	[0.77;1.52]	0.647
ratio 25.26 (group 19/ group 20)	1.51	[1.12;2.04]	0.007	1.27	[0.99;1.62]	0.063	1.14	[0.85;1.52]	0.376	1.65	[1.13;2.42]	0.01	1.01	[0.70;1.46]	0.969
ratio 3.3 (group 17/ group 18)	1.06	[0.78;1.44]	0.702	1.35	[0.99;1.86]	0.061	1.13	[0.79;1.61]	0.503	1.41	[0.95;2.09]	0.085	0.6	[0.36;1.00]	0.048
ratio 3.16 (group 11/ group 12)	1.23	[1.00;1.51]	0.055	1.21	[1.00;1.46]	0.056	1.09	[0.87;1.37]	0.464	1.1	[0.87;1.40]	0.43	1.18	[0.87;1.61]	0.285
ratio 7.8 (group 3/ group 4)	1.45	[1.11;1.88]	0.006	1.09	[0.86;1.38]	0.484	0.98	[0.75;1.27]	0.854	1.04	[0.78;1.39]	0.772	3.89	[2.10;7.21]	<0.001

Table ST4: Predictive value of microbiome community indices and compositional ratios of pediatric participants prior to HCT for infectious outcomes. Multiple cox regression model, adjusted for age, gender, conditioning regimen, is displayed as hazard ratio (HR) with 95% confidence interval (CI). P value is adjusted for multiple comparisons.

The pattern of the predictive baseline phylogenetic ratios for infection (see SF4) have very different distributions. These include differences in whether numerator is protective or predictive of infection (top two plots numerator protective, bottom two predictive) and prevalence of phylotypes within stools (e.g. common in Bacteremia ratio but rarely found in Bacterial ratio). However we can not from this analysis determine the relative weight of protective to predictive phylotypes are or the absolute values of these ratios play in the pathogenesis of these infections.

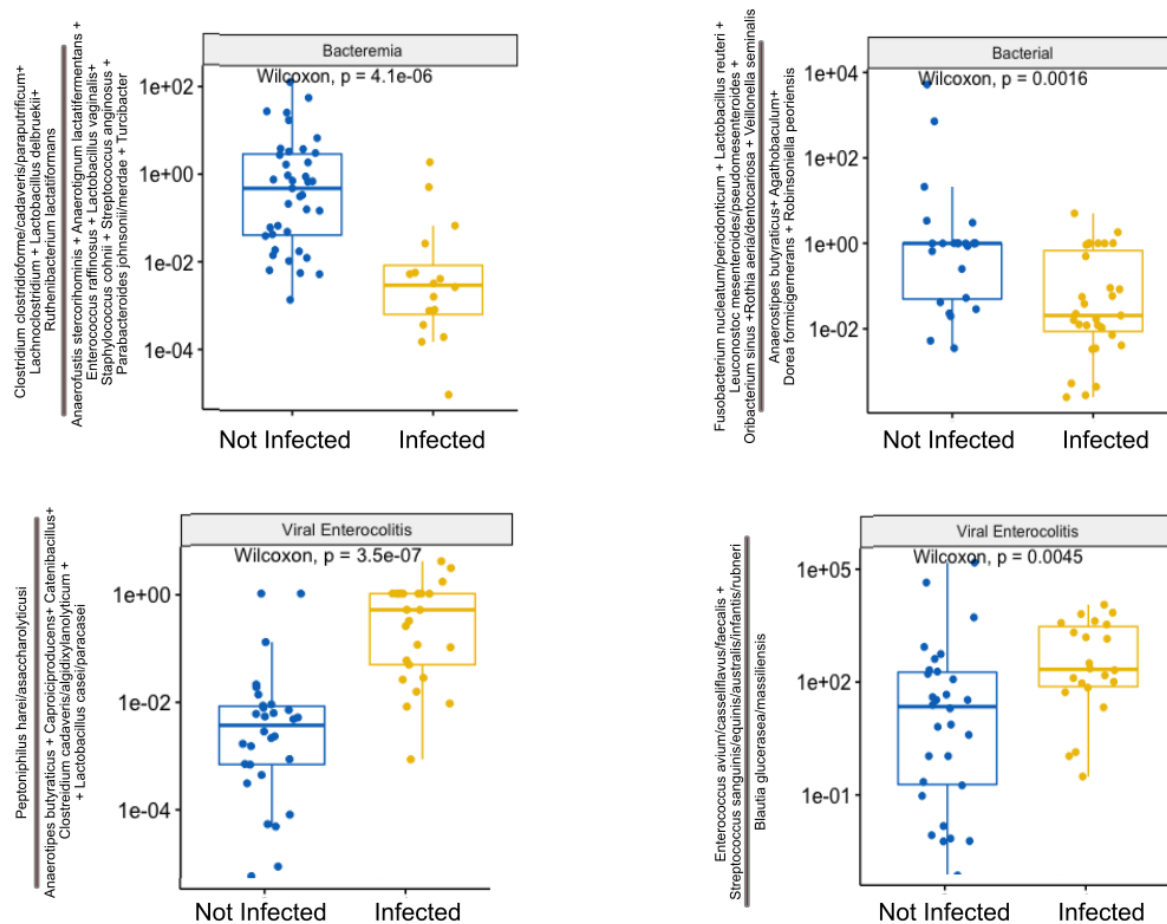


Figure SF4: Four ratios of bacteria taxa found in the microbiota prior HCT which differed in pediatric HCT participants who have specific infections within year after transplant then those that do not. The phylotypes (including a- group 3/group 4; b- group 1/group 2; c- group 5/group 6; and d- group 7/group 8) chosen for these ratios showed similar patterns of differing in abundance and distribution in different communities. Bacterial infections included bacteremias, pneumonias, UTIs and CDI. Viral infections included reactivation of systemic viruses, viral enterocolitis and viral LRTIs. P value is for Wilcoxon Signed Rank Test.

Neutrophil Recovery variable	Bacterial Infections			Viral Infections			Viral Enterocolitis			CDI			Bacteremia		
	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value
Bacterial Load (categorical 20th percentile)	1.32	[0.49;3.57]	0.588	1.73	[0.82;3.64]	0.15	1.22	[0.49;3.03]	0.667	0.29	[0.03;2.44]	0.254	1.17	[0.34;4.03]	0.803
Butyrate Producers (categorical 50th percentile)	0.72	[0.19;2.72]	0.630	0.35	[0.10;1.28]	0.113	0.35	[0.07;1.63]	0.180	0.63	[0.07;6.12]	0.691	0.83	[0.16;4.20]	0.819
Phylogenetic Diversity (categorical 20th percentile)	0.87	[0.28;2.69]	0.814	2.13	[0.85;5.33]	0.107	2.41	[0.78;7.45]	0.127	0.98	[0.15;6.18]	0.982	0.69	[0.15;3.17]	0.632
Single Dominant Taxa (categorical 90th percentile)	1.24	[0.38;4.00]	0.724	1.18	[0.49;2.87]	0.713	0.82	[0.30;2.25]	0.699	0.79	[0.14;4.42]	0.760	0.51	[0.11;2.42]	0.396
ratio 9.0 (group 10/ group 9)	1.2	[0.97;1.48]	0.096	1.33	[1.09;1.62]	0.006	1.35	[1.05;1.75]	0.019	0.96	[0.65;1.42]	0.843	1.2	[0.90;1.60]	0.224
ratio 12.0 (group 15/ group 16)	1.68	[1.11;2.55]	0.015	1.25	[0.92;1.69]	0.155	1.37	[0.93;2.03]	0.109	1.01	[0.56;1.82]	0.974	1.58	[0.89;2.82]	0.12
ratio 18.0 (group 7/ group 8)	0.99	[0.70;1.39]	0.947	1.15	[0.85;1.56]	0.357	1.12	[0.80;1.58]	0.505	0.98	[0.55;1.75]	0.936	0.77	[0.48;1.25]	0.298
ratio 5.0 (group 13/ group 14)	1.44	[1.09;1.89]	0.01	1.02	[0.81;1.29]	0.861	1.09	[0.84;1.42]	0.523	1.17	[0.75;1.83]	0.481	2.25	[1.33;3.78]	0.002
ratio 11.26 (group 1/ group 2)	0.89	[0.57;1.39]	0.601	0.99	[0.72;1.34]	0.925	1.01	[0.70;1.46]	0.972	0.91	[0.47;1.76]	0.777	0.69	[0.36;1.31]	0.26
ratio 16.16 (group 5/ group 6)	1.34	[0.87;2.07]	0.188	1	[0.74;1.36]	0.986	1.23	[0.85;1.77]	0.279	1.6	[0.76;3.38]	0.218	1.77	[0.87;3.59]	0.116
ratio 25.26 (group 19/ group 20)	0.71	[0.43;1.18]	0.189	1.04	[0.73;1.49]	0.829	0.94	[0.60;1.47]	0.785	0.56	[0.25;1.25]	0.16	0.59	[0.30;1.15]	0.119
ratio 3.3 (group 17/ group 18)	1.49	[0.99;2.23]	0.055	1.09	[0.80;1.50]	0.586	1.48	[1.01;2.18]	0.045	2.48	[1.35;4.58]	0.004	1.01	[0.59;1.73]	0.966
ratio 3.16 (group 11/ group 12)	1.81	[1.30;2.52]	<0.001	1.47	[1.11;1.94]	0.007	1.96	[1.38;2.79]	<0.001	1.73	[1.06;2.83]	0.028	1.52	[0.93;2.48]	0.096
ratio 7.8 (group 3/ group 4)	0.91	[0.66;1.25]	0.56	1.12	[0.88;1.42]	0.349	0.88	[0.65;1.20]	0.43	0.78	[0.51;1.22]	0.277	1.1	[0.73;1.68]	0.64

Table ST5: Predictive value of microbiome community indices and compositional ratios of sample at neutrophil recovery for infectious outcomes in pediatric HCT participants. Multiple cox regression model, adjusted for age, gender, conditioning regimen, is displayed as hazard ratio (HR) with 95% confidence interval (CI). P value is adjusted for multiple comparisons.

None of the phylotype ratios that were predictive of infection in the baseline (prior to HCT) microbiomes were predictive in the neutrophil recovery microbiomes. In fact the only ratio that was significant in both (ratio 5.0) was protective for bacterial infections at baseline and predictive of bacterial infections at neutrophil recovery. There was a single ratio that was able to predict whether participants had infections of any type (see SF5). This ratio suggests that the relative abundance of oral commensals was predictive when compared to strictly anaerobic intestinal commensals (protective). It is unclear whether the numerator (oral commensals) are triggers or signatures of disturbances or it was the loss of the phylotypes in the denominator (protective intestinal commensals), or both that influenced/associated with infections.

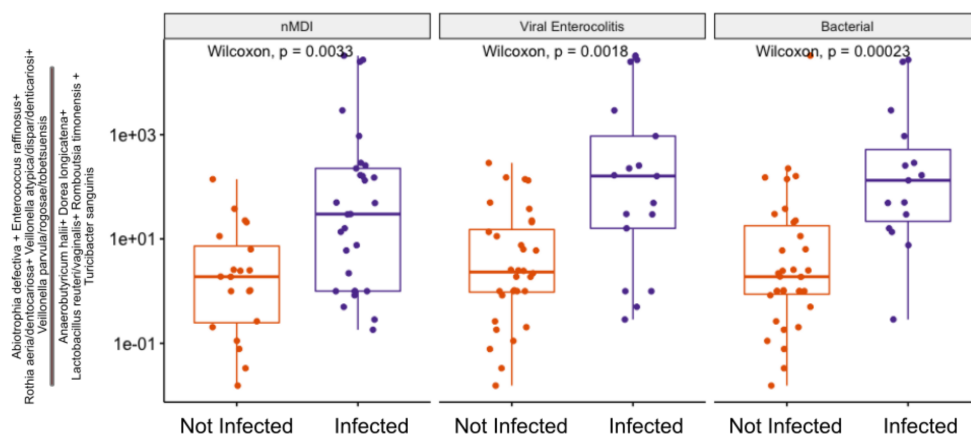


Figure SF5: A single ratio of bacteria taxa (group 11/ group 12) found in the microbiota at time of neutrophil recovery which differed in pediatric HCT participants who had infections or not. The phylotypes chosen for this ratio showed similar patterns of differing in abundance and distribution in different communities. Microbiologically defined infection (nMDI) included any bacterial, viral or parasitic infection. Bacterial infections included bacteremias, pneumonias, UTIs and CDI. P value is for Wilcoxon Signed Rank Test.

2.4 Cox Regression Analysis- Indices, Ratios and Acute GVHD Outcomes

microbiome variable	Prior to HCT			Neutrophil Recovery		
	HR	95% CI	P value	HR	95% CI	P value
Bacterial Load (categorical 20th percentile)	0.94	[0.43;2.08]	0.883	1.25	[0.55;2.88]	0.591
Butyrate Producers (categorical 50 percentile)	0.95	[0.35;2.54]	0.919	1.36	[0.46;4.00]	0.574
Phylogenetic Diversity (categorical 20th percentile)	0.47	[0.15;1.43]	0.184	2.38	[0.71;7.96]	0.158
Single Dominant Taxa (categorical 90th percentile)	0.72	[0.25;2.08]	0.541	1.01	[0.38;2.71]	0.987
ratio 9.0 (group 10/ group 9)	1.65	[1.22;2.23]	0.001	1.26	[0.99;1.60]	0.064
ratio 12.0 (group 15/ group 16)	1.23	[0.96;1.58]	0.107	1.61	[1.21;2.16]	0.001
ratio 18.0 (group 7/ group 8)	1.19	[0.92;1.54]	0.19	0.94	[0.71;1.25]	0.693
ratio 5.0 (group 13/ group 14)	0.93	[0.70;1.24]	0.638	1.15	[0.88;1.50]	0.301
ratio 11.26 (group 1/ group 2)	0.94	[0.70;1.25]	0.664	1.02	[0.70;1.49]	0.919
ratio 16.16 (group 5/ group 6)	1.11	[0.82;1.52]	0.497	0.94	[0.69;1.30]	0.727
ratio 25.26 (group 19/ group 20)	0.84	[0.62;1.14]	0.272	1.01	[0.68;1.51]	0.946
ratio 3.3 (group 17/ group 18)	0.97	[0.64;1.45]	0.87	0.94	[0.67;1.30]	0.689
ratio 3.16 (group 11/ group 12)	0.98	[0.76;1.27]	0.879	1.3	[0.94;1.81]	0.113
ratio 7.8 (group 3/ group 4)	1.45	[1.02;2.05]	0.036	1.02	[0.78;1.32]	0.905

Table ST6: Predictive value of microbiome community indices and compositional ratios of samples prior to HCT and at neutrophil recovery for acute GVHD (any grade) in pediatric HCT participants. Multiple cox regression model, adjusted for age, gender, graft type and ex vivo T cell depletion of graft, is displayed as hazard ratio (HR) with 95% confidence interval (CI). P value is adjusted for multiple comparisons.

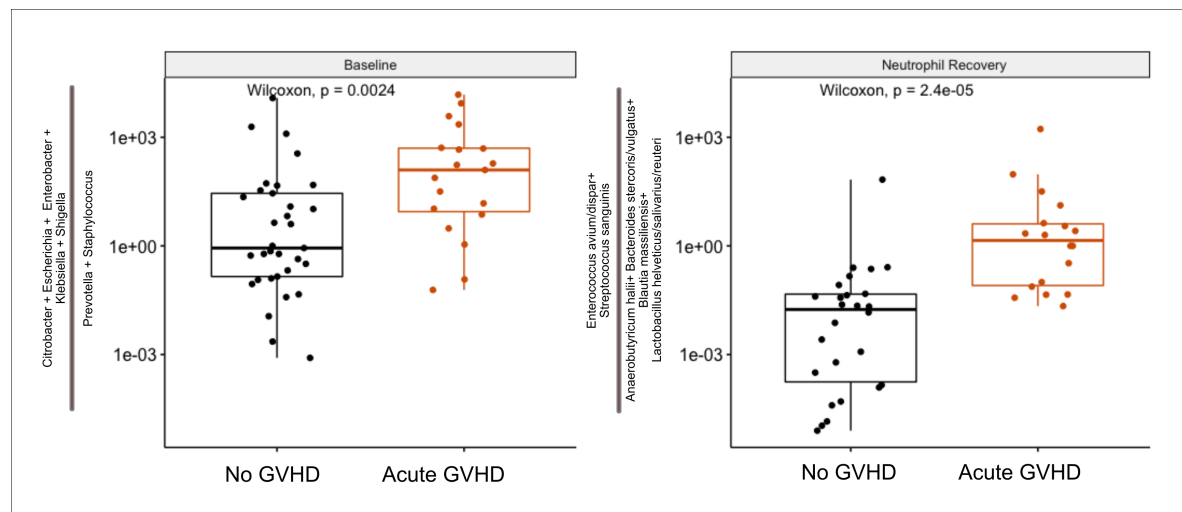


Figure SF6: Two ratios of bacteria taxa (a- group 10/group 9; b- group 15/group 16) found in the microbiota prior HCT or at neutrophil recovery which differed in pediatric HCT participants who have acute GVHD (any grade) within year after transplant. The phylotypes chosen for these ratios showed similar patterns of differing in abundance and distribution in different communities. P value is for Wilcoxon Signed Rank Test.

2.5 Cox Regression Analysis- Metabolic Pathways from Metagenomics

Category	ID	Metabolic Pathway Description	Presence in Phylotype Group's Genomes				Cox Regression Analysis			
			Numerator	Coverage	Denominator	Coverage	Outcome	Time Point	Hazards Ratio	95% CI
Lysine	PWY-5097	L-lysine biosynthesis VI	Group 1	37.6	Group 2	66.2	Bacterial	1	0.55	[0.29;1.03]
Anaerobic Nucleotide	PWY-7220 (PWY-7222)	adenosine/guanosine deoxyribonucleotides de novo biosynthesis II	Group 1	41	Group 2	0	CDI	1	1.39	[1.01;1.92]
LPS	OANTIGEN-PWY	O-antigen building blocks biosynthesis (E. coli)	Group 7	53.4	Group 8	0	Viral	1	1.79	[1.06;3.03]
Inositol	PWY-7237	myo-, chiro- and scyllo-inositol degradation	Group 7	6.1	Group 8	75.2	Viral	1	0.77	[0.60;1.00]
Arginine	PWY-7400	L-arginine biosynthesis IV (archaeobacteria)	Group 9	33.4	Group 10	51.7	aGVHD	1	8.17	[2.77;24.07]
Bile/Mucus/Pectin Degradation	GALACT-GLUCUROCAT-PWY	superpathway of hexuronide and hexuronate degradation	Group 9	0	Group 10	19.8	aGVHD	1	1.32	[1.11;1.56]
Anaerobic Heme	HEMESYN2-PWY	heme b biosynthesis II (oxygen-independent)	Group 9	0	Group 10	53.2	aGVHD	1	1.59	[1.22;2.07]
Arginine	PWY-5154	L-arginine biosynthesis III (via N-acetyl-L-citrulline)	Group 9	27.5	Group 10	48.9	aGVHD	1	2.63	[1.63;4.24]
Nucleotide recycling	PWY-6606	guanosine nucleotides degradation II	Group 9	0	Group 10	20	aGVHD	1	1.85	[1.34;2.54]
Lipid	PWY-6284	superpathway of unsaturated fatty acids biosynthesis (E. coli)	Group 9	0	Group 10	45.4	aGVHD	1	1.35	[1.05;1.73]
Autotrophs metabolism	PWY-5913	partial TCA cycle (obligate autotrophs)	Group 9	0	Group 10	40	aGVHD	1	1.45	[1.12;1.87]
Methane	METH-ACETATE-PWY	methanogenesis from acetate	Group 11	0	Group 12	9.7	Bacterial	2	1.23	[1.07;1.42]
Vitamin K	PWY-5837	2-carboxy-1,4-naphthoquinol biosynthesis	Group 11	31.2	Group 12	0	Bacterial	2	0.77	[0.65;0.92]
Vitamin K	PWY-5897	superpathway of menaquinol-11 (-12 or -13) biosynthesis	Group 11	15.3	Group 12	0	Bacterial	2	0.82	[0.69;0.98]
Phosphatidylglycerol	PWY4FS-7	phosphatidylglycerol biosynthesis I (plastidic)	Group 11	0	Group 12	11.1	Viral Enterocolitis	2	0.95	[0.80;1.13]
Anaerobic Hexose Catabolism	P124-PWY	Bifidobacterium shunt	Group 11	0	Group 12	12.1	Viral Enterocolitis	2	1.3	[1.11;1.52]
Vitamin K	PWY-5837	2-carboxy-1,4-naphthoquinol biosynthesis	Group 13	30.7	Group 14	4.6	Bacterial	2	0.77	[0.65;0.92]
Vitamin K	PWY-5897	superpathway of menaquinol-11 (-12 or -13) biosynthesis	Group 13	32.5	Group 14	0	Bacterial	2	0.82	[0.69;0.98]
Pyrimidine Salvage	PWY-7208	superpathway of pyrimidine nucleobases salvage	Group 15	35.2	Group 16	19.3	aGVHD	2	0.78	[0.67;0.92]

Table ST7: Compositional ratios that were predictive for infection and immunologic outcomes prior to HCT (time point 1) or at neutrophil recovery (time point 2) were assessed based on the absence or presence of metabolic pathways in their genomes. Coverage is calculated based on number of genomes in the group where the metabolic pathway was detected, with 100 being all genomes confidently detected metabolic pathway and 0 being not detected in any genomes. Intermediate coverage implies either not all genomes contained pathway or some genomes may contain pathway but detection was at a lower confidence score. For those pathways that differed between numerator and denominator of the compositional ratios we tested whether the normalized counts of that pathway in metagenomic of the stool samples was sufficient to predict the outcome (associated with the ratios). Multiple cox regression model, adjusted for age, gender, graft type and ex vivo T cell depletion of graft (for acute GVHD) and adjusted for age, gender, conditioning regimen (for infectious outcomes), is displayed as hazard ratio (HR) with 95% confidence interval (CI).

3 Phylotypes Included in Ratios

Below is a table of representative sequences from phylotypes which were used in constructing the phylotype ratios.

V3V4 Sequence	Phylotype	Classified Taxonomy	NCBI
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCGACGCCCGCGTGAAGGAAGAAGTATCTCGGTATGTAAACTTCTA TCAGCAGGGAAGAAAATGACGGTACCTGACTAAGAAGCCCCGGCTAACTACGTGCCAGCAGCCCGGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGAATAGCAAAGTCTGATGTGAAAGGCTGGGGCTTAACCC CAGGACTGCATTGGAACCTGTTGTTCTAGAGTGCCGAGAGGTAAGCGGAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACGGTAACCTGACGTTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	pt___00001	Blautia coccoides or Blautia hansenii	1532 or 1322
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCGACGCCCGCGTGAAGGAAGAAGTATCTCGGTATGTAAACTTCTA TCAGCAGGGAAGAAAATGACGGTACCTGACTAAGAAGCCCCGGCTAACTACGTGCCAGCAGCCCGGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGAAGAGCAAAGTCTGATGTGAAAGGCAGGGGCCCCAACCC CTGGACTCGCATGGAACCTGTCCTTCTTGAGTACCCGAGAGGTAAGCGGAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACGGTAACCTGACGTTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	pt___00001	Blautia stercoris or Blautia hansenii or Blautia coccoides	871664 or 1322 or 1532
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCGACGCCCGCGTGAAGGAAGAAGTATCTCGGTATGTAAACTTCTA TCAGCAGGGAAGAAAATGACGGTACCTGACTAAGAAGCCCCGGCTAACTACGTGCCAGCAGCCCGGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGAAGAGCAAAGTCTGATGTGAAAGGCAGGGGGCTCAACCC CCGGATGCATTGGAACCTGTTGTTCTTGAGTACCCGAGAGGTAAGCGGAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACGGTAACCTGACGTTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	pt___00001	Blautia stercoris or Blautia hansenii or Blautia hansenii	871664 or 1322 or 1322
TGGGGAATCTTCCGCAATGGACGAAAGTCTGACGGAGCAACGCCCGCGTGAGTGATGACGGCCCTTCGGGTTGTAAAGCTCTG TTAATCGGGACGAATGGTTCTTGTGCGAATAGTGGCAGGATTTGACGGTACCGGAATAGAAAGCCACGGCTAACTACGTGC CAGCAGCCGCGTAATACGTAGGTGGCAAGCGTTGTCCGGAATTTATGGGCGTAAAGCGCGCGCAGGCGGATCAGTTAGTC TGTCTTAAAGTTCCGGGGCTTAACCCCGTGATGGGATGGAAACTGCTGATCTAGAGTATCGGAGAGGAAAGTGGAATTCCT AGTGTAGCGGTGAAATGCGTAGATATTAGGAAGAACACCAGTGGCGAAGGCGACTTCTTGGACGAAAACCTGACGCTGAGGC GCGAAAGCCAGGGAGCGAACCGG	pt___00002	Veillonella atypica	39777
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TGGGGAATATTGCACAATGGGCGGAAGCCTGATGCAGCGACGCCCGCGTGAGGGATGGAGGCCCTTCGGGTTGTGAACCTCTT TCGCCCGTGGTCAAGCCGCAACTGTGGGTTGTGGTGAGGGTAGTGGGTAAAGAAGCGCCCGCTAACTACGTGCCAGCAGCC GCGGTAATACGTAGGGCGCAGCGTTGTCCGGAATTTATGGGCGTAAAGGGCTTGTAGGCGGCTGGTCCGCTCTGCCGTGA AATCCTCTGGCTCAACTGGGGCGTGCGGTGGGTACGGGCTGGCTTGAAGTGGGTAGGGGAGGCTGGAACCTCCTGGTGTAG CGGTGGAATGCGCAGATATCAGGAAGAACACCAGTGGCGAAGGCGGCTCTCTGGGCCGTACTGACGCTGAGGAGCGAAAG CGTGGGAGCGAACAGG	pt___00003	Schaalia georgiae	52768
TGGGGAATATTGCACAATGGGCGGAAGCCTGATGCAGCAACGCCCGCGTGAGCGATGAAGGCCCTTCGGGTCGTAAAGCTCTG TCCTCAAGGAAGATAATGACGGTACTTGAGGAGGAAGCCCCGGCTAACTACGTGCCAGCAGCCCGGGTAATACGTAGGGGG CTAGCGTTATCCGGAATTAAGTGGGCGTAAAGGGTGCGTAGGTGGTTTCTTAAGTCAGAAGTGAAAGGCTACGGCTCAACCG TAGTAAGCTTTTGAACCTGAGAACTTGAGTGACAGGAGAGGAGCGTAGAATTCCTAGTGTAGCGGTGAAATGCGTAGATAT TAGGAGGAATACCAGTTGCGAAGGCGGCTCTCTGGACTGTAACCTGACACTGAGGACGAAAGCGTGGGGAGCAAACAGG	pt___00005	Romboutsia sedi- mentorium	1368474
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TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTGTGAAGAAGGCCCTTCGGGTTGTAAAGCACTT TCAGCGGGGAGGAAGGCGTTAAGGTTAATAACCTTGGCGATTGACGTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCC AGCAGCCGCGGTAATACGGAGGGTGCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCG GATGTGAAATCCCCGGGCTTAACCTGGGAACATCGAAACTGGCAGGCTAGAGTCTTGTAGAGGGGGGTAGAATTCCA GGTGTAGCGGTGAAATGCGTAGATCTGGAGGAATACCGGTGGCGAAAGCGGCCCTGGACAAAGACTGACGCTCAGGT GCGAAAGCGTGGGGAGCAAACAGG	pt___00005	Klebsiella pneumo- niae	573
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TAGGGAATCTTCGGCAATGGACGAAAGTCTGACCGAGCAACGCCGCGGTGAGTGAAGAAGGTTTTCGGATCGTAAAACTCTG TTGTTAGAGAAGAACAAGGATGAGAGTAGAAGCTTCATCCCTTGACGGTATCTAACCAGAAAGCCACGGCTAACTACGTGCC AGCAGCCGCGGTAATACGTAGGTGGCAAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGGCGGTTTCTTAAGTCT GATGTGAAAGCCCCCGGCTCAACCGGGGAGGGTCATTGGAAACTGGGAGACTTGAGTGCAGAAGAGGAGAGTGGAATTCCA TGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCGAGTGGCGAAAGCGGCTCTCTGGTCTGTAACCTGACGCTGAGGC TCGAAAGCGTGGGGAGCAAACAGG	pt___00009	Enterococcus galli- narum	1353
TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTGTGAAGAAGGCCCTTCGGGTTGTAAAGCACTT TCAGCGGGGAGGAAGGCGGTGAGGTTAATAACCTCACCGATTGACGTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCC AGCAGCCGCGGTAATACGGAGGGTGCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCG GATGTGAAATCCCCGGGCTCAACCTGGGAACATCGAAACTGGCAGGCTAGAGTCTTGTAGAGGGGGGTAGAATTCCA GGTGTAGCGGTGAAATGCGTAGATCTGGAGGAATACCGGTGGCGAAAGCGGCCCTGGACAAAGACTGACGCTCAGGT GCGAAAGCGTGGGGAGCAAACAGG	pt___00009	Klebsiella variicola	244366
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCGACGCCGCGGTGAGCGAAGAAGTATTTCGGTATGTAAAGCTCTA TCAGCAGGGAAGAAAATGACGGTACCTGACTAAGAAGCCCCGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGCCAGGCAAGTCTGATGTGAAAGGCAGGGGCTCAACCC CTGGACTGCATTGGAACATGCCAGGCTGGAGTGCCGGAGAGGTAAGCGGAATTCCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCACTGGCGAAGGCGGCTTACTGGACGGTAACTGACGTTGATGCTCGAAAGCGTGGGGAGCAAACAGG	pt___00015	Murimonas intestini	1337051
TAGGGAATCTTCGGCAATGGGGGCAACCCCTGACCGAGCAACGCCGCGGTGAGTGAAGAAGGTTTTCGGATCGTAAAGCTCTG TTGTAAGTCAAGAACGGGTGTGAGAGTGGAAGTTACACTATGACGGTAGCTTACCAGAAAGGGACGGCTAACTACGTGC CAGCAGCCGCGGTAATACGTAGGTCCCGAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGGCGGTTTGATAAGTC TGAAGTTAAAGGCTGTGGCTCAACCATAGTTCCGCTTTGGAAACTGTCAAACCTTGAGTGCAGAAGGGGAGAGTGGAATTCCA TGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCGGTGGCGAAAGCGGCTCTCTGGTCTGTAACCTGACGCTGAGGC TCGAAAGCGTGGGGAGCGCAACAGG	pt___00016	Streptococcus sali- varius	1304
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCAACGCCGCGGTGAGTGATGACGGCCTTCGGGTTGTAAAGCTCTG TCTTCAGGGACGATAATACCGGTACCTGAGAAAGGAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTGG CGAGGTTGTCCGATTTACTGGGCGTAAAGGGAGCGTAGGCGGACTTTTAAAGTGAGATGTGAAATACCCGGGCTCAACTT GGGTGCTGCATTTCAAACCTGGAAGTCTAGAGTGCAGGAGAGGAGAATGGAATTCCCTAGTGTAGCGGTGAAATGCGTAGAGA TTAGGAAGAACACCACTGGCGAAGGCGATTCTCTGGACTGTAACCTGACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	pt___00016	Clostridium dis- poricum	84024

TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCCTTCGGGTTGTAAAGTACTTTCAGCGAGGAGGAAGGCGTTGTGGTTAATAACCGCAACGATTGACGTTACTCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGTGCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGATGTGTAATCCCCGGGCTCAACCTGGGAACTCGATCCGAAAACCTGGCAGGCTAGAGTCTTGTAGAGGGGGGTAGAATTCCA	pt__00018	Citrobacter freundii	546
GGTGTAGCGGTGAAATGCGTAGATATCTGGAGGAATACCGGTGGCGAAGGCGGCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAACAGG			
TAGGGAATCTTCGGCAATGGACGAAAGTCTGACCGAGCAACGCCGCGTGAGTGAAGAAGGTTTTCGGATCGTAAAACTCTGTTGTTAGAGAAGAACAAGGACGTTAGTAACTGAACGTCCCCTGACGGTATCTAACCAGAAAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTGGCAAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGGCGGTTTCTTAAGTCTGATGTGAAAGCCCCCGGCTCAACCGGGGAGGGTCATTGGAAAACCTGGGAAAACCTTGAGTGCAGAAGAGGAGAGTGGAAATTCATGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGCGGAAGGCGGCTCTCTGGTCTGTAACGTGACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAGG	pt__00022	Enterococcus faecalis	1351
TGAGGAATATTGGTCAATGGGCGATGGCCTGAACCAGCCAAGTAGCGTGAAGGATGACTGCCCTATGGGTTGTAACTTCTTTTATAAAGGAATAAAGTCGGGTATGCATACCCGTTTGCATGTACTTTATGAATAAGGATCGGCTAACTCCGTGCCAGCAGCGCGGTAATACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGATGGATGTTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAAATTGCAGTTGATACTGGATGCTTGAGTGCAGTTGAGGCAGGCGGAATTCGTGGTGTCGGGTGAAATGCTTAGATATCAGGAAGAACTCCGATTGCCAAGGCAGCGCTGCTAAGCTGCAACTGACATTGAGGCTCGAAA	pt__00022	Phocaeicola dorei	357276
GTGTGGGTATCAAACAGG			
TAGGGAATCTTCGGCAATGGGCGAAAGCCTGACCGAGCAACGCCGCGTGAAATGATGAAGGCCCTTCGGGTTGTAAAAATTCTGTTATAAAGGAATAAAGTCCTCTAGTAGGAAATGGCTAGAGTGTGACGGTACCTTATGAGAAAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTGGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGATGGATGTTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAAATTGCAGTTGATACTGGATGCTTGAGTGCAGTTGAGGCAGGCGGAATTCGTGGTGTCGGGTGAAATGCTTAGATATCAGGAAGAACTCCGATTGCCAAGGCAGCGCTGCTAAGCTGCAACTGACATTGAGGCTCGAAA	pt__00023	Turicibacter sanguinis	154288
GTGTGGGTATCAAACAGG			
TAGGGAATCTTCGGCAATGGGCGAAAGCCTGACCGAGCAACGCCGCGTGAAATGATGAAGGCCCTTCGGGTTGTAAAAATTCTGTTATAAAGGAATAAAGTCCTCTAGTAGGAAATGGCTAGAGTGTGACGGTACCTTATGAGAAAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTGGCGAGCGTTATCCGGATTTATTGGGCGTAAAGAGCGCGCAGGTGGTTGATTAAGTCTGATGTGAAAGCCCACGGCTTAACCGTGGAGGGTCATTGGAAAACCTGGTCAACTTGAGTGCAGAAGAGGGAAGTGGAATTCCA	pt__00025	Clostridium butyricum	1492
TGTGTAGCGGTGAAATGCGTAGATATGGAGGAACACCAGTGCGCAAGGCGGCTTCTGGTCTGTAACGTGACACTGAGGC			
GCGAAAGCGTGGGGAGCAAACAGG			
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCAACGCCGCGTGAGTGATGACGGTCTTCGGATTGTAAAGCTCTGTCTTTAGGGACGATAATGACGGTACCTAAGGAGGAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTGGCAAGCGTTGTCCGGATTTACTGGGCGTAAAGGGAGCGTAGGTGGATATTTAAGTGGGATGTGAAATACCCGGGCTTAACCTGGGTGCTGCATTCCAACTGGATATCTAGAGTGCAGGAGAGGAAAGGAGAAATTCCTAGTGTAGCGGTGAAATGCGTAGAGA	pt__00025	Enterocloster aldenensis	358742
TTAGGAGGAACACCAGTGCGCAAGGCGGCTTACTGGACTGTAACCTGACGTTGAGGCTCGAAAGCGTGGGGAGCAAACAGG			
TAGGGAATCTTCCGCAATGGGCGAAAGCCTGACGGAGCAACGCCGCGTGAGTGATGAAGGTCTTCGGATCGTAAAACTCTGTTATCAGGGAAGAACAACCGTGTAAAGTAACTGTGCACGTCTTGACGGTACCTGATCAGAAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTGGCAAGCGTTATCCGGATTTATTGGGCGTAAAGCGCGCGTAGGCGGTTTTTTAAGTCTGATGTGAAAGCCCACGGCTCAACCGTGGAGGGTCATTGGAAAACCTGGAAAACCTTGAGTGCAGAAGAGGAAAGTGGAATTCCATGTGTAGCGGTGAAATGCGCAGAGATATGGAGGAACACCAGTGCGCAAGGCGACTTTCTGGTCTGTAACCTGACGCTGATGTGCGAAAGCGTGGGGATCAAACAGG	pt__00027	Staphylococcus pasteurii	45972
TAGGGAATCTTCCGCAATGGGGGCAACCCCTGACCGAGCAACGCCGCGTGAGTGAAAGAAGGTTTTCGGATCGTAAAGCTCTGTTGTAAGTCAAGAACGGGTGTGAGAGTGGAAGTTTACACTGTGACGGTAGCTTACCAGAAAAGGGACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTCCCCGAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGGCGGTTTGATAAGTCTGAAGTTAAAGACTGTGGCTCAACCATAGTTCGCTTTGGAAAACCTGTCAAACCTTGAGTGCAGAAGGGGAGAGTGGAATTCCA	pt__00031	Streptococcus salivarius or Streptococcus vestibularis	1304 or 1343
TGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGCGCAAGGCGCTCTCTGGTCTGTAACCTGACGCTGAGGC			
TCCAAAGCGTGGGGAGCGAACAGG			
TAGGGAATCTTCCGCAATGGGGGCAACCCCTGACCGAGCAACGCCGCGTGAGTGAAAGAAGGTTTTCGGATCGTAAAGCTCTGTTGTAAGAGAAGAACGAGTGTGAGAGTGGAAGTTTACACTGTGACGGTATCTTACCAGAAAAGGGACGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGTCCCCGAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGGCGGTTAGATAAGTCTGAAGTTAAAGCTGTGGCTTAACCATAGTACGCTTTTGAAGCTGTTTAACTTGAGTGCAGAAGGGGAGAGTGGAATTCCA	pt__00033	Streptococcus	1301
TGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGCGCAAGGCGCTCTCTGGTCTGTAACCTGACGCTGAGGC			
TCCAAAGCGTGGGGAGCAAACAGG			

TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCGACGCCCGCTGAGGGATGGAGGCCCTTCGGGTTGTAAACCTCTT TTGTTAGGGAGCAAGGCACCTTTGTGTTGAGTGTACCTTTTGAATAAGCACCGGCTAACTACGTGCCAGCAGCCGCGGTAATA CGTAGGGTGAAGCGTTATCCGGAATTATTGGGCGTAAAGGGCTCGTAGGCGGTTTCGTCCGCTCCGGTGTGAAAGTCCATC GCTTAACGGTGGATCCGCGCCGGGTACGGGCGGGCTTGAGTGCGGTAGGGGAGACTGGAATTCCCGGTGTAAACGGTGGAAAT GTGTAGATATCCGGAAGAACACCAATGGCGAAGGCAGGTCTCTGGGCCGTTACTGACGCTGAGGAGCGAAAGCGTGGGGAG CGAACAGG	pt__00039	Bifidobacterium breve	1685
TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCGACGCCCGCTGAGGGATGGAGGCCCTTCGGGTTGTAAACCTCTT TTATCGGGGAGCAAGCGAGAGTGAAGTTTACCCGTTGAATAAGCACCGGCTAACTACGTGCCAGCAGCCGCGGTAATAACGTA GGGTGAAGCGTTATCCGGAATTATTGGGCGTAAAGGGCTCGTAGGCGGTTTCGTCCGCTCCGGTGTGAAAGTCCATCGCTT AACGGTGGATCCGCGCCGGGTACGGGCGGGCTTGAGTGCGGTAGGGGAGACTGGAATTCCCGGTGTAAACGGTGGAAATGTGT AGATATCCGGAAGAACACCAATGGCGAAGGCAGGTCTCTGGGCCGTTACTGACGCTGAGGAGCGAAAGCGTGGGGAGCGAA CAGG	pt__00039	Bifidobacterium longum	216816
TAGGGAATCTTCCACAATGGGCGCAAGCCTGATGCAGCAACCCGCTGAGTGAAGAAGGGTTTCGGCTCGTAAAGCTCTG TTGTTAAAGAAGAACAGTATGAGAGCAACTGTTTCATACGTTGACGGTATTTAACCAGAAAGTCAAGGCTAACTACGTGCCA GCAGCCGCGGTAATACGTAGGTGGCAAGCGTTATCCGGATTTATTGGGCGTAAAGAGAGTGCAGGCGGTTTTCTAAGTCTG ATGTGAAAGCCTTCGGCTTAACCGGAGAAGTGCATCGGAACTGGATAACTTGAGTGCAGAAGAGGGTAGTGGAACCTCCAT GTGTAGCGGTGGAATGCGTAGATATATGGAAGAACACCAAGTGGCGAAGGCGGCTACCTGGTCTGCAACTGACGCTGAGACT CGAAAGCATGGGTAGCGAACAGG	pt__00042	Limosilactobacillus fermentum	1613
TGAGGAATATTGGTCAATGGGCGCTAGCCTGAACCAGCCAAGTAGCGTGAAGGATGAAGGCTCTATGGGTGCTAAACTTCT TTTATATAAGAATAAAAGTGCAGTATGTATAGCTGTTTTGTATGTATTATATGAATAAGGATCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGACTGGTAAGTCAGTTGTG AAAGTTTGGCGCTCAACCGTAAAAATTGCAGTTGAAACTGGCAGTCTTGAGTACAGTAGAGGTGGGCGGAATTTCGTGGTGTA GCGGTGAATGCTTAGATATCACGAAGAACTCCGATTGCGAAGGCAGCTCACTAGACTGCAACTGACACTGATGCTCGAAAG TGTGGGTATCAAACAGG	pt__00042	Bacteroides fragilis	817
TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCGACGCCCGCTGAGGGATGGAGGCCCTTCGGGTTGTGAACTCTT TCGCCAGTGAAGCAGGCCCTGCCCTCGTTTGTGGGTGGGTTGACGGTAGCTGGATAAGAAGCGCCGGCTAACTACGTGCCAGC AGCCGCGGTAATACGTAGGGCGCGAGCGTTGTCCGGAATTATTGGGCGTAAAGAGCTCGTAGGCGGCTGGTCCGCTCTGTC GTGAAATCCTCTGGCTTAACCTGGGGGCTTGCGGTGGGTACGGGCCCGCTTGAGTGCGGTAGGGGAGACTGGAACCTCCTGGT GTAGCGGTGGAATGCGCAGATATCAGGAAGAACACCGGTGGCGAAGGCGGGTCTCTGGGCCGTTACTGACGCTGAGGAGCG AAAGCGTGGGGAGCGAACAGG	pt__00043	Actinomyces naes- lundii	1655
TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCGACGCCCGCTGAGGGATGGAGGCCCTTCGGGTTGTAAACCTCTT TCGCCAGTGAAGCAGGCCCTGCCCTCGTTTGTGGGTGGGTTGACGGTAGCTGGATAAGAAGCGCCGGCTAACTACGTGCCAGC AGCCGCGGTAATACGTAGGGCGCGAGCGTTGTCCGGAATTATTGGGCGTAAAGAGCTCGTAGGCGGCTGGTCCGCTCTGTC GTGAAATCCTCTGGCTTAACCTGGGGGCTTGCGGTGGGTACGGGCCCGCTTGAGTGCGGTAGGGGAGACTGGAACCTCCTGGT GTAGCGGTGGAATGCGCAGATATCAGGAAGAACACCGGTGGCGAAGGCGGGTCTCTGGGCCGTTACTGACGCTGAGGAGCG AAAGCGTGGGGAGCGAACAGG	pt__00043	Actinomyces naes- lundii or Actino- myces oris	1655 or 544580
TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCGACGCCCGCTGAGGGATGGAGGCCCTTCGGGTTGTAAACCTCTT TCGCCAGTGAAGCAGGCCCTGCCCTTTGTGGGTGGGTTGACGGTAGCTGGATAAGAAGCGCCGGCTAACTACGTGCCAGC AGCCGCGGTAATACGTAGGGCGCGAGCGTTGTCCGGAATTATTGGGCGTAAAGAGCTCGTAGGCGGCTGGTCCGCTCTGTC GTGAAATCCTCTGGCTTAACCTGGGGGCTTGCGGTGGGTACGGGCCCGCTTGAGTGCGGTAGGGGAGACTGGAACCTCCTGGT GTAGCGGTGGAATGCGCAGATATCAGGAAGAACACCGGTGGCGAAGGCGGGTCTCTGGGCCGTTACTGACGCTGAGGAGCG AAAGCGTGGGGAGCGAACAGG	pt__00043	Actinomyces oris	544580
TGAGGAATATTGGTCAATGGGCGAGAGCCTGAACCAGCCAAGTAGCGTGCAGGATGACGGCCCTATGGGTTGTAAACTGCT TTTGTATGGGATAAAGTCAGTCAGTGTGATTGTTTGCAGGTACCATACGAATAAGGACCGGCTAATTCCGTGCCAGCAGC CGCGGTAATACGGAAGGTCCGGGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGCTGGAGATTAAGTGTGTTGTG AAATGTAGACGCTCAACGCTGACTTGCAGCGCATACTGGTTTCCTTGAGTACGCACAACGTTGGCGGAATTTCGTGCTGTAG CGGTGAAATGCTTAGATATGACGAAGAACTCCGATTGCGAAGGCAGCTGACGGGAGCGCAACTGACGCTGAAGCTCGAAGG TGCGGGTATCGAACAGG	pt__00046	Prevotella histicola	470565
TGGGGAATATTGCACAATGGGGGAAACCCCTGATGCAGCGACGCCCGCTGAGTGAAGAAGTATCTCGGTATGTAAAGCTCTA TCAGCAGGAAAGAAAATGACGGTACCTGACTAAGAAGCCCGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGGG CAAGCGTTATCCGGAATTACTGGGTGTAAAGGGTGCGTAGGTGGTATGGCAAGTCAGAAGTGAAGAACCCAGGGCTTAACCTC TGGGACTGCTTTTGAAGCTGTCAGACTGGAGTGCAAGGAGAGGTAAGCGGAATTTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACATCAGTGGCGAAGGCGGCTTACTGGACTGAAACTGACACTGAGGCACGAAAGCGTGGGGAGCAAACAGG	pt__00047	Anaerostipes hadrus	649756

TGGGGGATATTGCACAATGGGGGAAACCCTGATGCAGCAACGCCCGCTGAGGGGAAGAAGGTTTTCGGATTGTAAACCTCTG TTCTTAGTGACGATAATGACGGTAGCTAAGGAGAAAAGCTCCGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGAG CGAGCGTTGTCCGGATTTACTGGGTGTAAAGGGTGCGTAGGCGGCGAGGCAAGTCAGGCGTGAAATCTATGGGCTTAACCC ATAAACTGCGCTTGAACCTGTCTTGCTTGAGTGAAGTAGAGCTAGGCGGAATTTCCCGGTGTAGCGGTGAAATGCGTAGAGA TCGGGAGGAACACCAAGTGGCGAAGGCCGGCTACTGGGCTTTAACTGACGCTGAAGCACGAAAGCATGGGTAGCAAACAGG TAGGGAATCTTTCGGCAATGGACGAAAGTCTGACCGAGCAACGCCCGCTGAGTGAAGAAGGTTTTCGGATCGTAAACCTCTG TTGTTAGAGAAGAACAAGGATGAGAGTAACTGTTTCATCCCTTGACGGTATCTAACCAGAAAGCCACGGCTAACTACGTGCCA GCAGCCGCGTAATACGTAGGTGGCAAGCGTTGTCCGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTTCTTAAGTCTG ATGTGAAAGCCCCCGGCTCAACCGGGAGGGTCATTGGAACCTGGGAGACTTGAGTGCAGAAGAGGAGAGTGGAATTCCAT GTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGGCGAAGGCGACTCTCTGGTCTGTAACCTGACGCTGAGGCT CGAAAGCGTGGGGAGCAAACAGG	pt__00053	[Clostridium] tum	lep-	1535
TAGGGAATCTTTCGGCAATGGACGAAAGTCTGACCGAGCAACGCCCGCTGAGTGAAGAAGGTTTTCGGATCGTAAACCTCTG TTGTTAGAGAAGAACAAGGATGAGAGTAACTGTTTCATCCCTTGACGGTATCTAACCAGAAAGCCACGGCTAACTACGTGCCA GCAGCCGCGTAATACGTAGGTGGCAAGCGTTGTCCGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTTCTTAAGTCTG ATGTGAAAGCCCCCGGCTCAACCGGGAGGGTCATTGGAACCTGGGAGACTTGAGTGCAGAAGAGGAGAGTGGAATTCCAT GTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGGCGAAGGCGACTCTCTGGTCTGTAACCTGACGCTGAGGCT CGAAAGCGTGGGGAGCAAACAGG	pt__00054	Enterococcus	thai- landicus	417368
TGGGGAATATTGCGCAATGGGGGAAACCCTGACGCAGCAACGCCCGCTGAGCGATGAAGGTTTTCGGATCGTAAAGCTCTG TCCTTGGGGGAAGATAATGACGGTACCCCAAGGAGGAAGCTCCGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGAG CAAGCGTTGTCCGGATTACTGGGCGTAAAGAGCACGTAAGCGGTTAAATTAAGTCAGGTGTGAAAAGTTTTCGGCTCAACCG GAAAAGTGCACCTTGAACCTGAATAACTTGAGTATCGGAGAGGTAAGCGGAATTCCTAGTGTAGCGGTGAAATGCGTAGAGA TTAGGAAGAACAACCGGTGGCGAAGGCCGGCTTACTGGACGATAAAGTACGCGTGAAGTGCAGAAAGCGTGGGGAGCGAACAGG TAGGGAATCTTTCGGCAATGGGGGCAACCCTGACCGAGCAACGCCCGCTGAGTGAAGAAGGTTTTCGGATCGTAAAGCTCTG TTGTAAGAGAAGAACGAGTGTGAGAGTGGAAAGTTTACGCTGTGACGGTATCTTACCAGAAAAGGACGGCTAACTACGTGC CAGCAGCCGCGGTAATACGTAGGTCCCGAGCGTTATCCGGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTTAGATAAGTC TGAAGTTAAAGGCTGTGGCTTAACCATAGTACGCTTTGGAAACTGTTTAACTTGAGTGCAAGAGGGGAGAGTGGAATTCCA TGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGGCGAAGCGGCTCTCTGGCTGTAACTGACGCTGAGGC TCGAAAGCGTGGGGAGCAAACAGG	pt__00055	Anaerofustis	ster- cori-hominis	214853
TAGGGAATCTTTCGGCAATGGGGGCAACCCTGACCGAGCAACGCCCGCTGAGTGAAGAAGGTTTTCGGATCGTAAAGCTCTG TTGTAAGAGAAGAACGAGTGTGAGAGTGGAAAGTTTACGCTGTGACGGTATCTTACCAGAAAAGGACGGCTAACTACGTGC CAGCAGCCGCGGTAATACGTAGGTCCCGAGCGTTATCCGGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTTAGATAAGTC TGAAGTTAAAGGCTGTGGCTTAACCATAGTACGCTTTGGAAACTGTTTAACTTGAGTGCAAGAGGGGAGAGTGGAATTCCA TGTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGGCGAAGCGGCTCTCTGGCTGTAACTGACGCTGAGGC TCGAAAGCGTGGGGAGCAAACAGG	pt__00055	Streptococcus	per- oris	68891
TGGGGAATATTGCACAATGGGGGAAAGCCTGATGCAGCAACGCCCGCTGAGCGATGAAGGCCCTTCGGGTGCTAAAGCTCTG TCCTCAAGGAAGATAATGACGGTACTTGAGGAGGAAGCCCCGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGGG CTAGCGTTATCCGGAATTACTGGGCGTAAAGGGTGCGTAGGCGGCTCTTCAAGTCAGGAGTGAAAGGCTACGGCTCAACCG TAGTAAGCTCTTGAACTGTAAGACTTGAGTGTAGGAGAGGAGAGTGAATTCCTAGTGTAGCGGTGAAATGCGTAGATAT TAGGAGGAATACCAGTTGCGAAGGCCGGCTCTCTGGACTGTAACCTGACGCTGAGGCACGAAAGCGTGGGGAGCAAACAGG TGAGGAATATTGGTCAATGGCCGAGAGGCTGAACCGCAAGTCCGCTGAAGGATGAAGGATCTATGGTTTGTAAACTTCT TTTATAGGGGAATAAAGTGTGGGACGTGTTCCATTTTGTATGTACCCTATGAATAAGCATCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACCGAGGATGCGAGCGTTATCCGGATTATTTGGGTTTAAAGGGTGCGTAGGTGGTAATTTAAGTCAGCGGTG AAAGTTTGTGGCTCAACCATAAAATTGCCGTGGAACCTGGGTTACTTGAGTGTGTTTGAGGTAGGCGGAATGCGTGGTGTA GCGGTGAAATGCATAGATATCACGCAGAACTCCAATTGCGAAGGCAGCTTACTAAACCATAAAGTACACTGAAGCACGAAAG CGTGGGTATCAAACAGG	pt__00057	Peptostreptococcaceae		186804
TGAGGAATATTGGTCAATGGCCGAGAGGCTGAACCGCAAGTCCGCTGAAGGAAGAAGGATCTATGGTTTGTAAACTTCT TTTATAGGGGAATAAAGTGTGGGACGTGTCCTTTTGTATGTACCCTATGAATAAGCATCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACCGAGGATGCGAGCGTTATCCGGATTATTTGGGTTTAAAGGGTGCGTAGGTGGTAATTTAAGTCAGCGGTG AAAGTTTGTGGCTCAACCATAAAATTGCCGTGGAACCTGGGTTACTTGAGTGTGTTTGAGGTAGGCGGAATGCGTGGTGTA GCGGTGAAATGCATAGATATCACGCAGAACTCCGATTGCGAAGGCAGCTTACTAAACCATAAAGTACACTGAAGCACGAAAG CGTGGGTATCAAACAGG	pt__00067	Parabacteroides johnsonii		387661
TGAGGAATATTGGTCAATGGCCGAGAGGCTGAACCGCAAGTCCGCTGAAGGAAGAAGGATCTATGGTTTGTAAACTTCT TTTATAGGGGAATAAAGTGTGGGACGTGTCCTTTTGTATGTACCCTATGAATAAGCATCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACCGAGGATGCGAGCGTTATCCGGATTATTTGGGTTTAAAGGGTGCGTAGGTGGTAATTTAAGTCAGCGGTG AAAGTTTGTGGCTCAACCATAAAATTGCCGTGGAACCTGGGTTACTTGAGTGTGTTTGAGGTAGGCGGAATGCGTGGTGTA GCGGTGAAATGCATAGATATCACGCAGAACTCCGATTGCGAAGGCAGCTTACTAAACCATAAAGTACACTGAAGCACGAAAG CGTGGGGATCAAACAGG	pt__00067	Parabacteroides merdae		46503
TGGGGAATATTGCACAATGGAGGAAACTCTGATGCAGCGACGCCCGCTGAAGGATGAAGTATTTTCGGTATGTAAACTTCTA TCAGCAGGGAAGAAAATGACGGTACCTGACTAAGAAGCCCCGGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGCACGGCAAGCCAGATGTGAAAAGCCCGGGGCTCAACCC CGGACTGCATTTGGAACCTGCTGAGCTAGAGTGTCCGAGAGGCAAGTGGAAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAAGTGGCGAAGGCCGGCTTGCTGGACGATGACTGACGTTGAGGCTCGAAAGCGTGGGGAGCAAACAGG TAGGGAATCTTCCGCAATGGACGCAAGTCTGACGGAGCAACGCCCGCTGAGTGAAGAAGGATTTTCGGTTTCGTAAAACTCTG TTGTTAGAGAAGAAGACAGCGCATAGAGTAACTGTTATGCGTGTGACGGTATCTAACCAGAAAGCCACGGCTAACTACGTGCCA GCAGCCGCGTAATACGTAGGTGGCAAGCGTTGTCCGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTCATTAAGTCTG ATGTGAAAGCCCCCGGCTCAACCGGGGAGGGTCATTGGAACCTGGTTGACCTTGAGTGCAGAAGAGGAGAGTGGAATTCCAT GTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGGCGAAGGCGACTCTCTGGTCTGTAACCTGACGCTGAGGCT CGAAAGCGTGGGTAGCAAACAGG	pt__00068	Dorea longicatena		88431
TAGGGAATCTTCCACAATGGACGAAAGTCTGATGGAGCAACGCCCGCTGAGTGAAGAAGGTTTTCGGATCGTAAAGCTCTG TTGTTGGTGAAGAAGGATAGAGGTAGTAACTGGCTTTTATTTGACGGTAATCAACCAGAAAGTCACGGCTAACTACGTGCCA GCAGCCGCGTAATACGTAGGTGGCAAGCGTTGTCCGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTAATTAAGTCTG ATGTGAAAGCCCCCGGCTCAACCGGGGAGGGTCATTGGAACCTGGTTGACCTTGAGTGCAGAAGAGGAGAGTGGAATTCCAT GTGTAGCGGTGAAATGCGTAGATATATGGAGGAACACCAGTGGCGAAGGCGACTCTCTGGTCTGTAACCTGACGCTGAGGCT CGAAAGCGTGGGTAGCAAACAGG	pt__00074	Granulicatella elegans		137732
TAGGGAATCTTCCACAATGGACGAAAGTCTGATGGAGCAACGCCCGCTGAGTGAAGAAGGTTTTCGGATCGTAAAGCTCTG TTGTTGGTGAAGAAGGATAGAGGTAGTAACTGGCTTTTATTTGACGGTAATCAACCAGAAAGTCACGGCTAACTACGTGCCA GCAGCCGCGTAATACGTAGGTGGCAAGCGTTGTCCGATTATTTGGGCGTAAAGCGAGCGCAGGCGGTAATTAAGTCTG ATGTGAAAGCCCCCGGCTCAACCGAGGAACTGCATCGGAAACTGTTTTTCTTGAGTGCAGAAGAGGAGAGTGGAATTCCAT GTGTAGCGGTGAAATGCGTAGATATATGGAAGAACACCAGTGGCGAAGGCGACTCTCTGGTCTGCAACTGACGCTGAGGCT CGAAAGCATGGGTAGCGAACAGG	pt__00074	Lactobacillus	hel- veticus	1587

TGAGGAATATTGGTCAATGGACGCAAGTCTGAACCAGCCATGCCGCGTGCAGGAAGACGGCTCTATGAGTTGTAAACTGCT TTTGTACGAGGGTAACAGCATCTACGCGTAGGTGCATGAAAGTATCGTACGAATAAGGATCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACGGAGGATCCAAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCCTAGGCGGTTTATAAGTTAGAGGTT AAATATCGGAGCTTAACTCCGTTATGCCTCTAATACTGTAGGACTAGAGAATAGTTGCGGTAGGCGGAATGTATGGTGTAGC GGTGAATGCTTAGAGATCATACAGAACCCGATTGCGAAGGCAGCTTACCAAACATATGACTGACGTTGAGGCACGAAAGC GTGGGGAGCAAACAGG	pt__00097	Alistipes	239759
GGGGAATATTGGGCAATGGGCGCAAGCCTGACCCAGCAACGCCGCGTGAAGGAAGAAGGCTTTCGGGTTGTAAACTTCTTT TGTCAGGGACGAAACAAATGACGGTACCTGACGAATAAGCCACGGCTAACTACGTGCCAGCAGCCGCGATAAATACGTAGGG GGCTAGCGTTATCCGGAATTACTGGGCGTAAAGGGTGCCTAGGCGGTCTTTCAAGCCAGAAAGTAAAAGGCTACGGCTCAAC CGTAGTAAGCTTTTGGAACTGTAGGACTTGAGTGCAGGAGAGGAGAGTGGAATTCCTAGTGTAGCGGTGAAATGCGTAGAT ATTAGGAGGAACACCAGTAGCGAAGGCGGCTCTCTGGACTGTAACGTACGCTGAGGCACGAAAGCGTGGGGAGCAAACAGG TGGGGAATATTGCACAATGGGCGAAAGCCTGATGCAGCGACGCCGCGTGAAGGAAGAAGTATTTCCGGTATGTAAAGCTCTA TCAGCAGGGAAGATAATGACAGTACCTGACTAAGAAGCCCCGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGGTGGCTAGGTAAGTCAGGTGTGAAAGCCCCGGGGCTCAACCC CGGGATTGCACCTGAAACTACTTAGCTAGAGTGCAGGAGAGGTAAGTGAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACTGTAACGTGACACTGAGGCTCGAAAGCGTGGGGAGCAAACAGG TGGGGAATATTGCACAATGGGCGAAACCCCTGATGCAGCGACGCCGCGTGAAGGAAGAAGTATTTCCGGTATGTAAAGCTCTA TCAGCAGGGAAGAAGAAATGACGGTACCTGACTAAGAAGCACC GGCTAAATACGTGCCAGCAGCCGCGGTAATACGTATGG TGCAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGCAGGCGGAAGGCTAAGTCTGATGTGAAAGCCCCGGGGCTCAAC CCCCGTTAGCTGCAATGGTTCATCTAGAGTGTGCGGAGGGTAAAGTGAATTCCTAGTGTAGCGGTGAAATGCGTAGA TATTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACGATAACTGACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAG G	pt__00097	Paeniclostridium ghonii	29358
TGGGGAATATTGCACAATGGGCGAAAGCCTGATGCAGCGACGCCGCGTGAAGGAAGAAGTATTTCCGGTATGTAAAGCTCTA TCAGCAGGGAAGATAATGACAGTACCTGACTAAGAAGCCCCGCTAACTACGTGCCAGCAGCCGCGGTAATACGTAGGGGG CAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGGTGGCTAGGTAAGTCAGGTGTGAAAGCCCCGGGGCTCAACCC CGGGATTGCACCTGAAACTACTTAGCTAGAGTGCAGGAGAGGTAAGTGAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACTGTAACGTGACACTGAGGCTCGAAAGCGTGGGGAGCAAACAGG TGGGGAATATTGCACAATGGGCGAAACCCCTGATGCAGCGACGCCGCGTGAAGGAAGAAGTATTTCCGGTATGTAAAGCTCTA TCAGCAGGGAAGAAGAAATGACGGTACCTGACTAAGAAGCACC GGCTAAATACGTGCCAGCAGCCGCGGTAATACGTATGG TGCAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGCAGGCGGAAGGCTAAGTCTGATGTGAAAGCCCCGGGGCTCAAC CCCCGTTAGCTGCAATGGTTCATCTAGAGTGTGCGGAGGGTAAAGTGAATTCCTAGTGTAGCGGTGAAATGCGTAGA TATTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACGATAACTGACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAG G	pt__00111	Mobilitalea sibirica or [Clostridium] fimetarium	1462919 or 99656
TGGGGAATATTGCACAATGGGCGAAACCCCTGATGCAGCGACGCCGCGTGAAGGAAGAAGTATTTCCGGTATGTAAAGCTCTA TCAGCAGGGAAGAAGAAATGACGGTACCTGACTAAGAAGCACC GGCTAAATACGTGCCAGCAGCCGCGGTAATACGTATGG TGCAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGCAGGCGGAAGGCTAAGTCTGATGTGAAAGCCCCGGGGCTCAAC CCCCGTTAGCTGCAATGGTTCATCTAGAGTGTGCGGAGGGTAAAGTGAATTCCTAGTGTAGCGGTGAAATGCGTAGA TATTAGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGACGATAACTGACGCTGAGGCTCGAAAGCGTGGGGAGCAAACAG G	pt__00117	Roseburia inulinivo- rans	360807
TGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCCTTCGGGTTGTAAAGTACTT TCAGTCCGGAGGAAGGTGTCAAGGTTAATAACCTTGGCAATTGACGTTACCGACAGAAGAAGCACC GGCTAACTCCGTGCC AGCAGCCGCGGTAATACGGAGGGTGCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTTGATTGAGTCA GATGTGAAATCCCCGGGCTTAAACCGGGAATTGCATCTGATACTGGTCAGCTAGAGTCTTTGTAGAGGGGGGTAGAATTCCA TGTGTAGCGGTGAAATGCGTAGAGATGTGGAGGAATACCGGTGGCGAAGGCGGCCCTTGACAAAGACTGACGCTCAGGT GCGAAAGCGTGGGGAGCAAACAGG	pt__00131	Morganella mor- gani	582
TAGGGAATCTTTTCAAAATGGGCGAAAGCCTGATGGAGCAATGCCGCGTGCAGGACGAAGGCCCTTCGGGTTGTAAACTGCTT TTAAAGCCGAGAAATATGATGGTAAGCTTTGAATAAGGATCGGCTAACTACGTGCCAGCAGCCGCGGTCATACGTAGGATCC GAGCATTATCCGGAGTGACTGGGTGTAAAGAGTTGCGTAGGTGGCAAGGTAAGTAGATAGTGAATCTGGTGGCTCAACCA TTCAGACTATTATCTAAACTATCTAGCTCGAGACTGTTATGGGTAACTGGAATTTCTAGTGTAGGAGTGAAATCCGTAGATA TTAGAAGGAACACCAATAGCGTAGGCAGGTTACTGGAACAGTTCTGACACTAAGGCACGAAAGCGTAGGGAGCAAACGGG TGAGGAATATTGGTCAATGGACGGAAGTCTGAACCAGCCAAAGTACGCTGCAGGATGACGGCCCTATGGGTTGTAAACTGCT TTTGTATGGGGATAAAGTTAGGGACGTGTCCCTATTTGCAGGTACCATACGAATAAGGACCCGCTAATTCCGTGCCAGCAGC CGCGGTAATACGGAAGGTCCAGGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGCTGGATATTAAGTGTGTTGTG AAATGTAGACGCTCAACGCTGACTTGCAGCGCATACTGGTTTTCTTGTAGTACGCACAACGTTGGCGGAATTCGTGCTGTAG CGGTGAAATGCTTAGATATGACGAAGAATCCGATTGCGAAGGCAGCTGACGGGAGCGCAACTGACGCTTAAGCTCGAAGG TGCGGGTATCAAACAGG	pt__00131	Candidatus Saccha- ribacteria	95818
TGAGGAATATTGGTCAATGGACGCAAGTCTGAACCAGCCATGCCGCGTGCAGGATGAAGGTGCTATGCATTGTAAACTGCT TTTGTACGAGGGTAAATGCAGGTACGTGTACCTGTTTGAAGATATCGTACGAATAAGGGTCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACGGAGGACCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCCTAGGCGGATTAGTAAGTTAGAGGTG AAAGCTCGATGCTCAACATCGAAATTGCCCTGATACTGTTAGTCTAGAGTATAGTTGCGGAAGGCGGAATGTGTGGTGTGTA CGGTGAAATGCTTAGATATCACACAGAACACCGATTGCGAAGGCAGCTTTCGAAGCTATTACTGACGCTGATGCACGAAAG CGTGGGGAGCGAACAGG	pt__00142	Prevotella	838
TGAGGAATATTGGTCAATGGACGCAAGTCTGAACCAGCCATGCCGCGTGCAGGATGAAGGTGCTATGCATTGTAAACTGCT TTTGTACGAGGGTAAATGCAGGTACGTGTACCTGTTTGAAGATATCGTACGAATAAGGGTCGGCTAACTCCGTGCCAGCAGC CGCGGTAATACGGAGGACCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCCTAGGCGGATTAGTAAGTTAGAGGTG AAAGCTCGATGCTCAACATCGAAATTGCCCTGATACTGTTAGTCTAGAGTATAGTTGCGGAAGGCGGAATGTGTGGTGTGTA CGGTGAAATGCTTAGATATCACACAGAACACCGATTGCGAAGGCAGCTTTCGAAGCTATTACTGACGCTGATGCACGAAAG CGTGGGGAGCGAACAGG	pt__00150	Alistipes indistinc- tus	626932
TAGGGAATCTTTCCACAATGGACGCAAGTCTGATGGAGCAACGCCGCGTGAAGGAAGAAGTCTTCGGATCGTAAAGCTCTG TTGTTGGTGAAGAAGGATAGAGGCAGTAAGTGTCTTTATTGTCAGTAATCAACCAGAAAGTACGCGCTAACTACGTGCCA GCAGCCGCGGTAATACGTAGGTGGCAAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGCGCAGCGGAATGATAAGTCTG ATGTGAAAGCCCACGGCTCAACCGTGGAAGTGCATCGGAAAGTGTATTCTTGAAGTGCAGAAAGAGGAGAGTGGAAGTCCAT GTGTAGCGGTGGAATGCTAGATATATGGAAGAACACCAGTGGCGAAGGCGGCTCTCTGGTCTGCAACTGACGCTGAGGCT CGAAAGCATGGGTAGCGAACAGG	pt__00157	Lactobacillus del- brueckii	1584
TGGGGAATATTGCACAATGGAGGAAACTCTGATGCAGCGACGCCGCGTGAAGGAAGAAGTCTTCGGATTGTAAACCTCTG TCTTCAGGAGGATAATGACGGTACCTGAGGAGGAAGCCACGGCTAACTACGTGCCAGCAGCCGCGGTAACACGTAGGTGG CAAGCGTTGTCCGGAATTACTGGGTGTAAAGGGAGTGCAGGCGGGACGGCAAGTTGGAAGTGAACCCCATGGGCTTAACCC ATGAAGTCTTTCAAACTGTCTTCTTGAAGTGGTGCAGAGGTAGGCGGAATTCGCCGTTGTAGCGGTGGAATGCGTAGATA TCGGGAGGAACACCAGTGGCGAAGGCGGCTTACTGGGCACTAACTGACGCTGAGGCTCGAAAGCATGGGTAGCAAACAGG	pt__00228	Eubacteriales	186802

TGGGGAATATTGCACAATGGGGGAAACCCTGATGCAGCAACGCCGCGTGAGTGAAGAAGTATTTTCGGTATGTAAAGCTCTA TCAGCAGGGAAGATAATGACGGTACCTGACTAAGAAGCTCCGGCTAAATACGTGCCAGCAGCCGCGGTAATACGTATGGAG CAAGCGTTATCCGAATTTACTGGGTGTAAAGGGTGCGTAGGTGGCAGTGCAAGTCAGATGTGAAAGGCCGGGGCTCAACCC CGGAGCTGCATTTGAAACTGCATAGCTAGAGTACAGGAGAGGCAGGCGGAATTCCTAGTGTAGCGGTGAAATGCGTAGATA TTAGGAGGAACACCAGTGGCGAAGGCGGCCCTGCTGGACTGTTACTGACACTGAGGCACGAAAGCGTGGGGAGCAAACAGG	pt__00421	Lachnospiraceae	186803
TGGGGAATATTGGACAATGGGGGAAACCCTGATCCAGCCATGCCGCGTGTGTGAAGAAGGCCCTTTTGGTTGTAAAGCACTT TAAGCGAGGAGGAGGCTTACCTGGTTAATACCTGGGATAAGTGGACGTTACTCGCAGAATAAGCACCGGCTAACTCTGTGC CAGCAGCCGCGGTAATACAGAGGGTGCAAGCGTTAATCGGATTTACTGGGCGTAAAGCGCGCGTAGGTGGCTAATTAAGTC AAATGTGAAATCCCCGAGCTTAACTTGGGAATTGCATTCGATACTGGTTAGCTAGAGTATGGGAGAGGATGGTAGAATTCC AGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGATGGCGAAGGCAGCCATCTGGCCTAATACTGACACTGAGG TGCGAAAGCATGGGGAGCAAACAGG	pt__00542	Acinetobacter vari- abilis or Acinetobac- ter boissieri	70346 or 1219383

Table ST8: Sequences of V3-V4 segment of rRNA gene that defined particular phylotypes included in ratios to describe compositional attributes with classified taxonomy and NCBI taxonomy identification. Phylogenetic classifications that could have been placed near two taxonomies were described with 'or' designation.

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